Module 6 Final Project - Final Report

ALY6040: Data Mining

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1 Introduction

The dataset for this final project contains detailed information on gun violence in the United States from January 2013 to March 2018. The data can help identify trends and patterns in gun violence. By analyzing the data, I hope to use data mining techniques to find important insights that may also help provide effective recommendations to promote a safer society.

Here are my business questions from Module 3:

1. What factors predict severe incidents?

2. Are there times of the year when incidents are more likely?

3. Which incidents are most likely to have multiple casualties?

4. How do local factors relate to high incident rates in certain states?

To answer these questions, I used a dataset containing details about gun violence incidents, including participant roles, locations, and characteristics. This analysis involves cleaning the data, exploring trends, and building models to predict high-casualty incidents.

2 Analysis

2.1 Data Preparation

2.1.1 Data Cleaning:

Removed missing and unneeded columns to make the data consistent.

2.1.2 Feature Engineering:

I combined the number of kills and injuries into a new feature called `total casualties` to represent the severity of each incident, and created a binary variable `high_casualty` to identify incidents with three or more casualties. I also use one-hot encoding to convert some non-numeric columns into `0` or `1`.

2.1.3 Encoding:

Converted categorical features like `state`, `month`, and `weekday` into numbers to work with machine learning models.

2.2 Methodologies

2.2.1 Tools:

I used R and several libraries like dplyr, ggplot2, and randomForest.

2.2.2 Techniques:

Exploratory Data Analysis (EDA): I looked at the most common ages, genders, and time trand involved in gun violence as well as the word cloud with the most common location.
Machine Learning Model: I used a `Random Forest` classifier to predict high-casualty incidents. This model helps to understand which features are most important in determining

severe cases. (I created Logistic Regression, LightGBM, and K-Nearest Neighbors models, but due to the long processing times and unsatisfactory results, I removed the code and kept only a summary of the final results in Model Comparison part.)

3 R Code for the Whole Gun Violence Analysis

3.1 Code for EDA

```r
library(tidyverse)
library(lubridate)
library(gridExtra)
library(ggrepel)
library(maps)
library(scales)
library(splitstackshape)

# Loading the dataset as a dataframe
gun_data <- read.csv("C:/Users/akbak/Downloads/gun-violence-data_01-2013_03-2018.csv")

# Summary of the dataset
summary(gun_data)
```

```
##   incident_id          date              state           city_or_county
##  Min.   :  92114   Length:239677      Length:239677      Length:239677
##  1st Qu.: 308545   Class :character   Class :character   Class :character
##  Median : 543587   Mode  :character   Mode  :character   Mode  :character
##  Mean   : 559334
##  3rd Qu.: 817228
##  Max.   :1083472
```

```
##
##    address             n_killed         n_injured      incident_url
##  Length:239677      Min.   : 0.0000   Min.   : 0.000   Length:239677
##  Class :character   1st Qu.: 0.0000   1st Qu.: 0.000   Class :character
##  Mode  :character   Median : 0.0000   Median : 0.000   Mode  :character
##                     Mean   : 0.2523   Mean   : 0.494
##                     3rd Qu.: 0.0000   3rd Qu.: 1.000
##                     Max.   :50.0000   Max.   :53.000
##
##   source_url        incident_url_fields_missing congressional_district
##  Length:239677      Length:239677               Min.   : 0.000
##  Class :character   Class :character            1st Qu.: 2.000
##  Mode  :character   Mode  :character            Median : 5.000
##                                                 Mean   : 8.001
##                                                 3rd Qu.:10.000
##                                                 Max.   :53.000
##                                                 NA's   :11944
##   gun_stolen          gun_type         incident_characteristics    latitude
##  Length:239677      Length:239677      Length:239677            Min.   :19.11
##  Class :character   Class :character   Class :character         1st Qu.:33.90
##  Mode  :character   Mode  :character   Mode  :character         Median :38.57
##                                                                  Mean   :37.55
##                                                                  3rd Qu.:41.44
##                                                                  Max.   :71.34
##                                                                  NA's   :7923
##  location_description   longitude       n_guns_involved     notes
##  Length:239677        Min.   :-171.43   Min.   :  1.00   Length:239677
##  Class :character     1st Qu.: -94.16   1st Qu.:  1.00   Class :character
##  Mode  :character     Median : -86.25   Median :  1.00   Mode  :character
##                       Mean   : -89.34   Mean   :  1.37
```

```
##                       3rd Qu.: -80.05   3rd Qu.:  1.00
##                       Max.   :  97.43   Max.   :400.00
##                       NA's   :7923      NA's   :99451
##  participant_age    participant_age_group participant_gender participant_name
##  Length:239677      Length:239677         Length:239677      Length:239677
##  Class :character   Class :character      Class :character   Class :character
##  Mode  :character   Mode  :character      Mode  :character   Mode  :character
##
##
##
##
##  participant_relationship participant_status participant_type
##  Length:239677            Length:239677      Length:239677
##  Class :character         Class :character   Class :character
##  Mode  :character         Mode  :character   Mode  :character
##
##
##
##
##    sources          state_house_district state_senate_district
##  Length:239677      Min.   :  1.00       Min.   : 1.00
##  Class :character   1st Qu.: 21.00       1st Qu.: 9.00
##  Mode  :character   Median : 47.00       Median :19.00
##                     Mean   : 55.45       Mean   :20.48
##                     3rd Qu.: 84.00       3rd Qu.:30.00
##                     Max.   :901.00       Max.   :94.00
##                     NA's   :38772        NA's   :32335
```

```r
# Checking data types of each column
data_types <- sapply(gun_data, class)
```

```r
data_types_df <- tibble(Column = names(data_types), DataType = data_types)

# Counting unique values in each column
unique_counts <- sapply(gun_data, function(x) length(unique(x)))
unique_counts_df <- tibble(Column = names(unique_counts), UniqueValues = unique_counts)

# Combining data types and unique counts
data_summary <- left_join(data_types_df, unique_counts_df, by = "Column") %>%
  arrange(UniqueValues)
data_summary
```

```
## # A tibble: 29 x 3
##    Column                      DataType  UniqueValues
##    <chr>                       <chr>            <int>
##  1 incident_url_fields_missing character            1
##  2 n_killed                    integer             16
##  3 n_injured                   integer             23
##  4 state                       character           51
##  5 congressional_district      integer             55
##  6 state_senate_district       integer             69
##  7 n_guns_involved             integer            107
##  8 participant_type            character          260
##  9 state_house_district        integer            276
## 10 participant_relationship    character          285
## # i 19 more rows
```

```r
# Finding and handling missing data
missing_data <- sapply(gun_data, function(x) sum(is.na(x)))
missing_percentage <- (missing_data / nrow(gun_data)) * 100
missing_summary <- tibble(
```

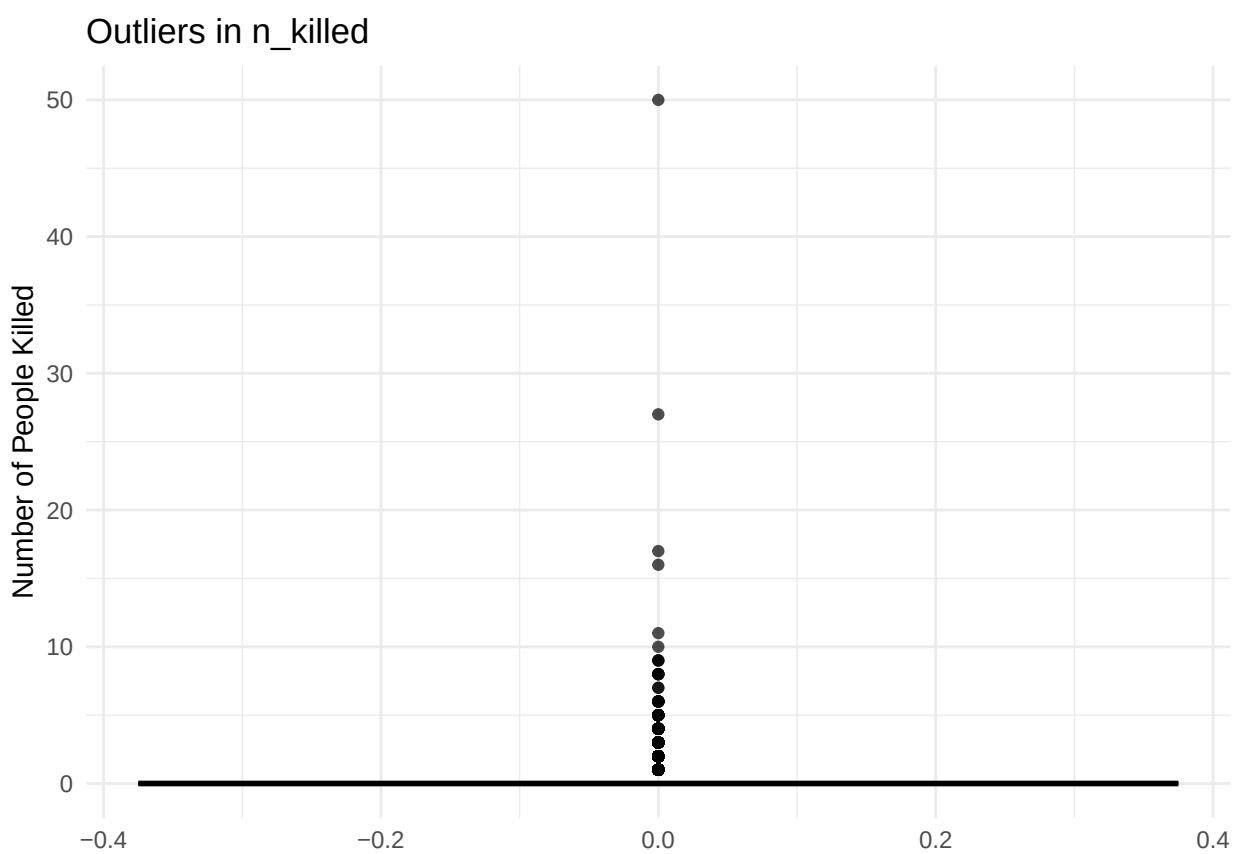
```r
  Column = names(missing_data),
  MissingCount = missing_data,
  MissingPercentage = missing_percentage
) %>%
  arrange(desc(MissingPercentage))
missing_summary
```

```
## # A tibble: 29 x 3
##    Column                 MissingCount MissingPercentage
##    <chr>                         <int>             <dbl>
##  1 n_guns_involved               99451             41.5
##  2 state_house_district          38772             16.2
##  3 state_senate_district         32335             13.5
##  4 congressional_district        11944              4.98
##  5 latitude                       7923              3.31
##  6 longitude                      7923              3.31
##  7 incident_id                       0              0
##  8 date                              0              0
##  9 state                             0              0
## 10 city_or_county                    0              0
## # i 19 more rows
```

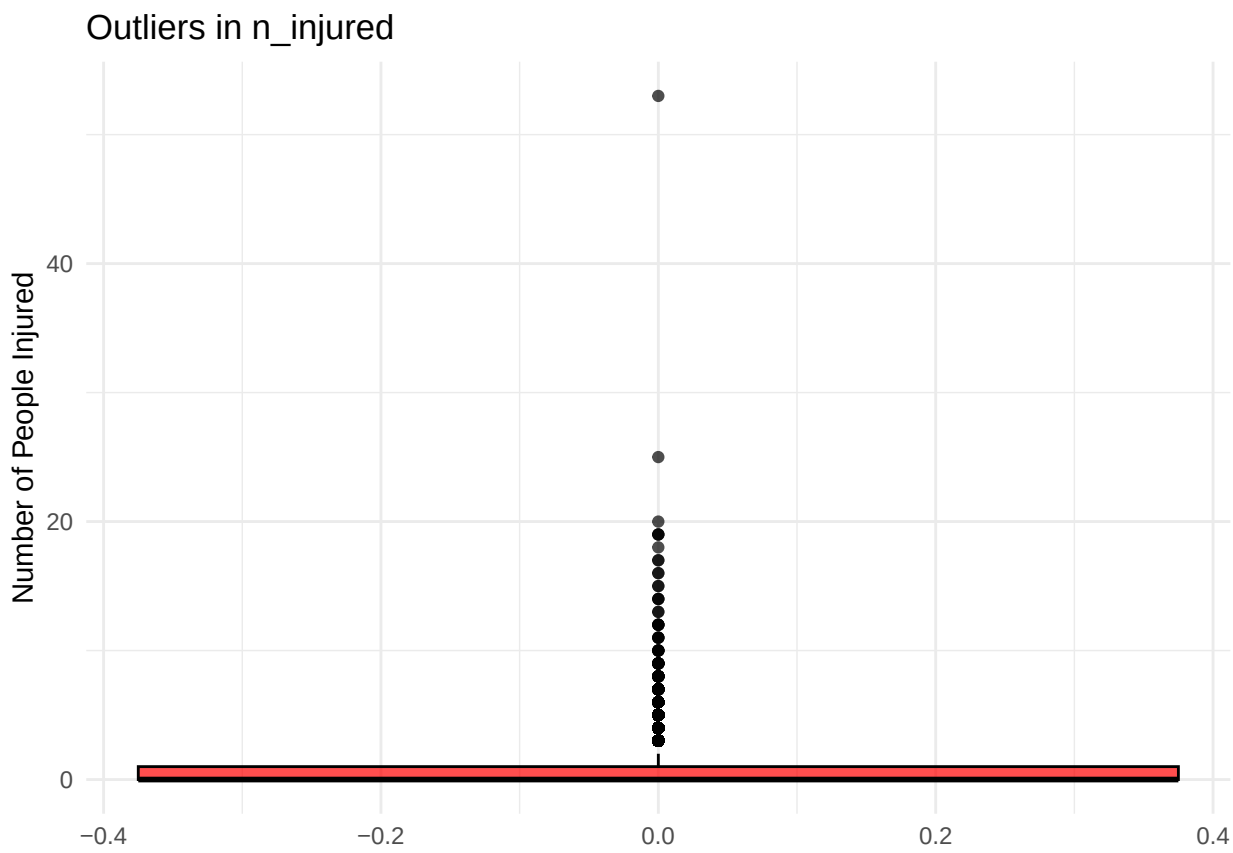
```r
# Detecting and removing duplicates
duplicate_count <- sum(duplicated(gun_data))
cat("Number of duplicate rows:", duplicate_count, "\n")
```

```
## Number of duplicate rows: 0
```

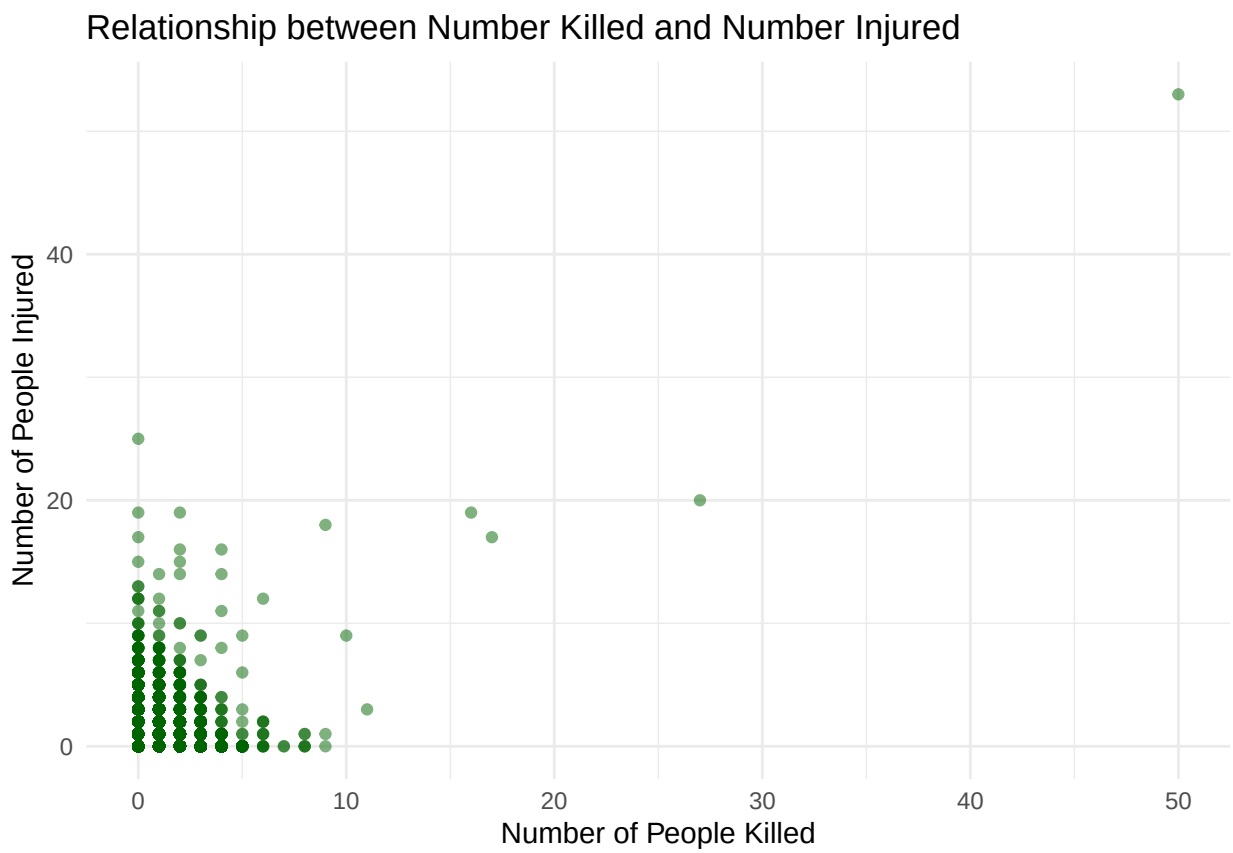
```r
# Visualizing outliers in 'n_killed' and 'n_injured'
ggplot(gun_data, aes(y = n_killed)) +
  geom_boxplot(fill = "blue", color = "black", alpha = 0.7) +
  labs(title = "Outliers in n_killed", y = "Number of People Killed") +
  theme_minimal()
```




```r
ggplot(gun_data, aes(y = n_injured)) +
  geom_boxplot(fill = "red", color = "black", alpha = 0.7) +
  labs(title = "Outliers in n_injured", y = "Number of People Injured") +
  theme_minimal()
```




```r
# Identifying relationships between 'n_killed' and 'n_injured'
ggplot(gun_data, aes(x = n_killed, y = n_injured)) +
  geom_point(alpha = 0.5, color = "darkgreen") +
  labs(title = "Relationship between Number Killed and Number Injured", x = "Number of P
  theme_minimal()
```



3.1.1 Data Preparation Summary

1. Data Types and Unique Values: Each column's data type and unique values were checked to aid analysis.

2. Handling Missing Data: High missing values in columns like `n_guns_involved` may impact modeling.

3. Duplicate Data: No duplicate rows found, indicating clean data.

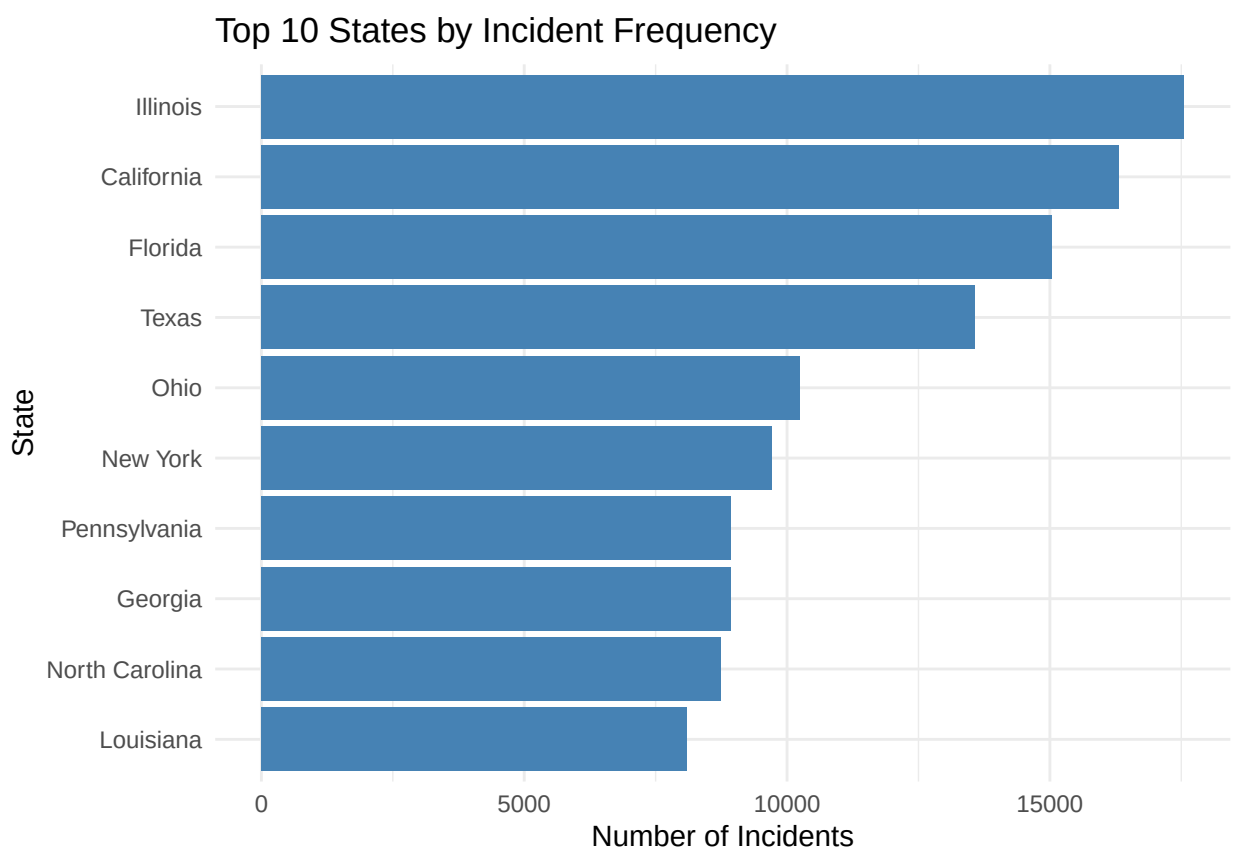
Summary: These steps ensure data quality and provide a foundation for further analysis and modeling.

4. `n_killed` and `n_injured`: Most incidents have low casualties, but a few show extreme high numbers of deaths and injuries.

5. Relationship Between `n_killed` and `n_injured` Correlation: There's a positive correlation between deaths and injuries, suggesting combining them as `total casualties` for a

better measure of severity.

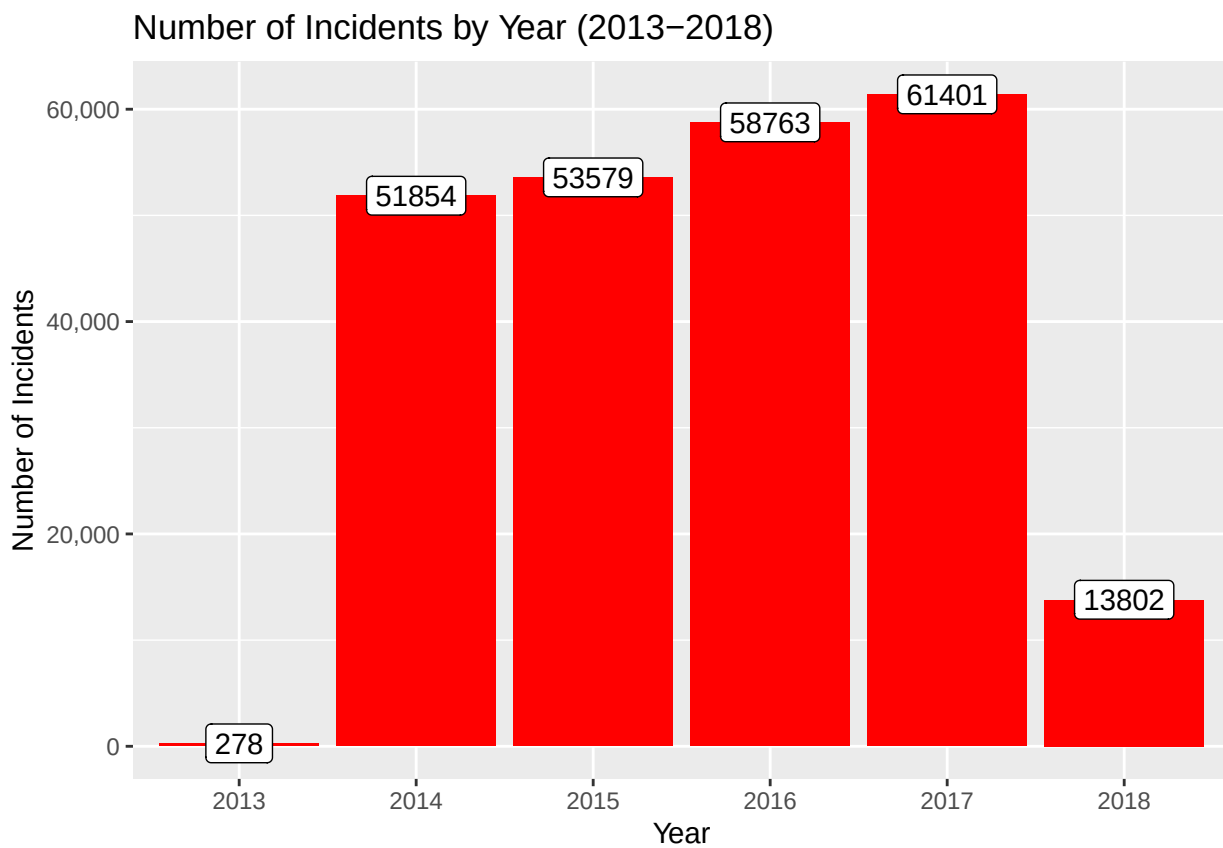
```
# Bar chart for top 10 states by incident frequency
top_states <- gun_data %>%
  count(state, sort = TRUE) %>%
  slice_max(n, n = 10)

ggplot(top_states, aes(x = reorder(state, n), y = n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Top 10 States by Incident Frequency", x = "State", y = "Number of Incide
  theme_minimal()
```



3.1.2 Insight:

Illinois, California, and Florida have the highest gun violence incident frequencies, indicating these states could benefit most from focused interventions.

```
# Extracting year and quarter
gun_data$date <- ymd(gun_data$date)
gun_data <- gun_data %>% mutate(year = year(date), quarter = quarter(date))

# Plotting number of incidents by year
gun_data %>%
  ggplot(aes(x = as.factor(year))) +
  geom_bar(stat = 'count', fill = 'red') +
  scale_y_continuous(labels = comma) +
  geom_label(stat = "count", aes(label = after_stat(count), y = after_stat(count))) +
  labs(x = 'Year', y = 'Number of Incidents', title = 'Number of Incidents by Year (2013
```

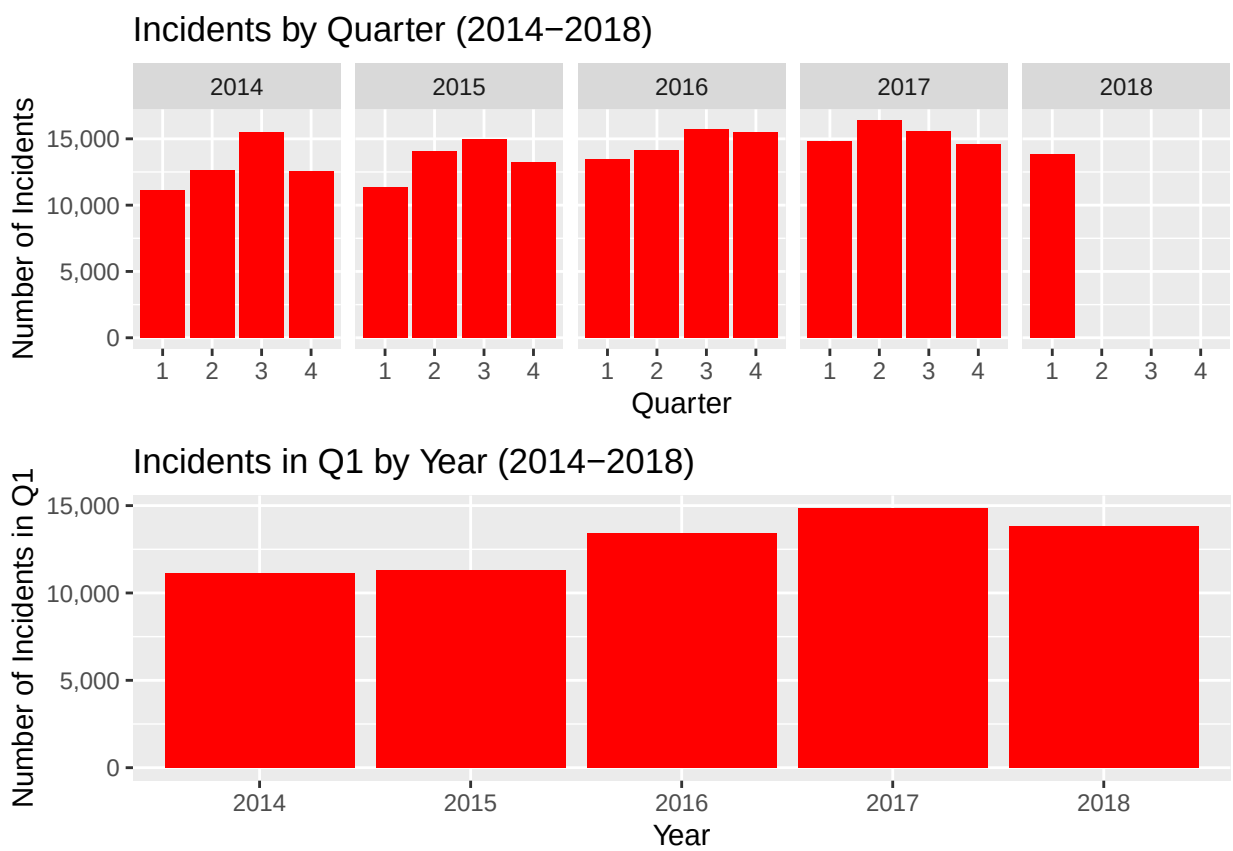


3.1.3 Insight:

Gun violence incidents increased yearly from 2014 to 2017, peaking in 2017, indicating a rising trend that might reflect worsening conditions or reporting improvements.

```
# Plotting number of incidents by quarter each year, excluding 2013
q1 <- gun_data %>%
  filter(year != 2013) %>%
  count(year, quarter) %>%
  ggplot(aes(x = as.factor(quarter), y = n)) +
  geom_col(fill = 'red') +
  scale_y_continuous(labels = comma) +
  facet_grid(. ~ year) +
  labs(x = 'Quarter', y = 'Number of Incidents', title = 'Incidents by Quarter (2014-201
```

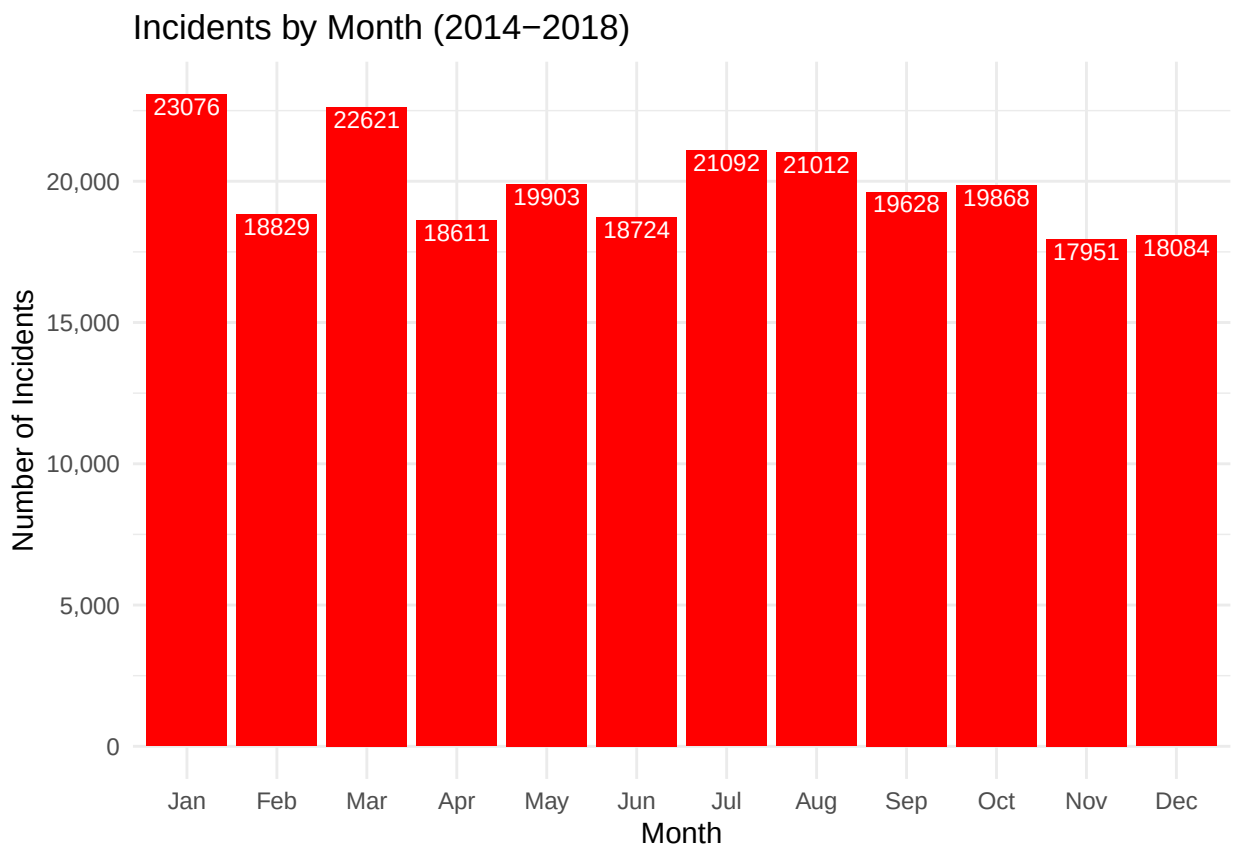
```r
# Plotting number of incidents in Q1 of each year, excluding 2013
q2 <- gun_data %>%
  filter(year != 2013 & quarter == 1) %>%
  count(year) %>%
  ggplot(aes(x = as.factor(year), y = n)) +
  geom_col(fill = 'red') +
  scale_y_continuous(labels = comma) +
  labs(x = 'Year', y = 'Number of Incidents in Q1', title = 'Incidents in Q1 by Year (20

# Displaying the plots side by side
grid.arrange(q1, q2, ncol = 1)
```



3.1.4 Insight:

Incident frequencies vary by quarter, with `Q3` generally seeing more incidents. Additionally, incidents increased each year from 2014 to 2017.

```r
# Extracting month
gun_data <- gun_data %>% mutate(month = month(date, label = TRUE))

# Plotting incidents by month
gun_data %>%
  filter(year %in% 2014:2018) %>%
  count(month) %>%
  ggplot(aes(x = month, y = n)) +
  geom_col(fill = 'red') +
  scale_y_continuous(labels = comma) +
  geom_text(aes(label = n), vjust = 1.2, color = "white", size = 3) +
  labs(x = 'Month', y = 'Number of Incidents', title = 'Incidents by Month (2014-2018)')
  theme_minimal()
```

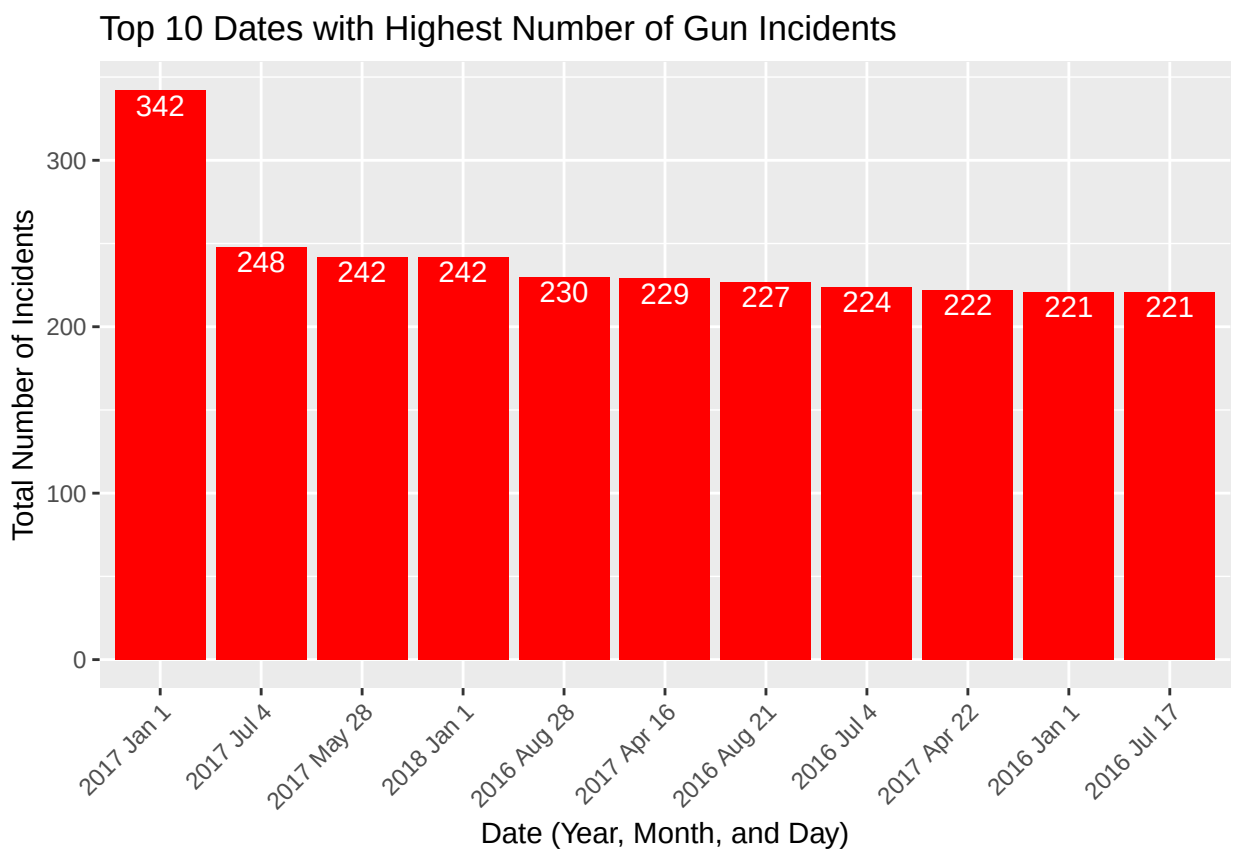


3.1.5 Insight:

`January` and `February` have the highest incident counts, suggesting early-year peaks in gun violence, possibly influenced by seasonal or situational factors.

```
# Extracting day from date
gun_data <- gun_data %>% mutate(day = day(date), date2 = str_c(year, " ", month, " ", da

# Plotting top 10 most frequent dates with incidents
gun_data %>%
  filter(year %in% 2014:2018) %>%
  count(date2, sort = TRUE) %>%
  slice_max(n, n = 10) %>%
  ggplot(aes(x = reorder(date2, -n), y = n)) +
```

```
  geom_col(fill = 'red') +
  geom_text(aes(label = n), vjust = 1.2, color = "white", size = 4) +
  scale_y_continuous(labels = comma) +
  labs(x = 'Date (Year, Month, and Day)', y = 'Total Number of Incidents', title = 'Top
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

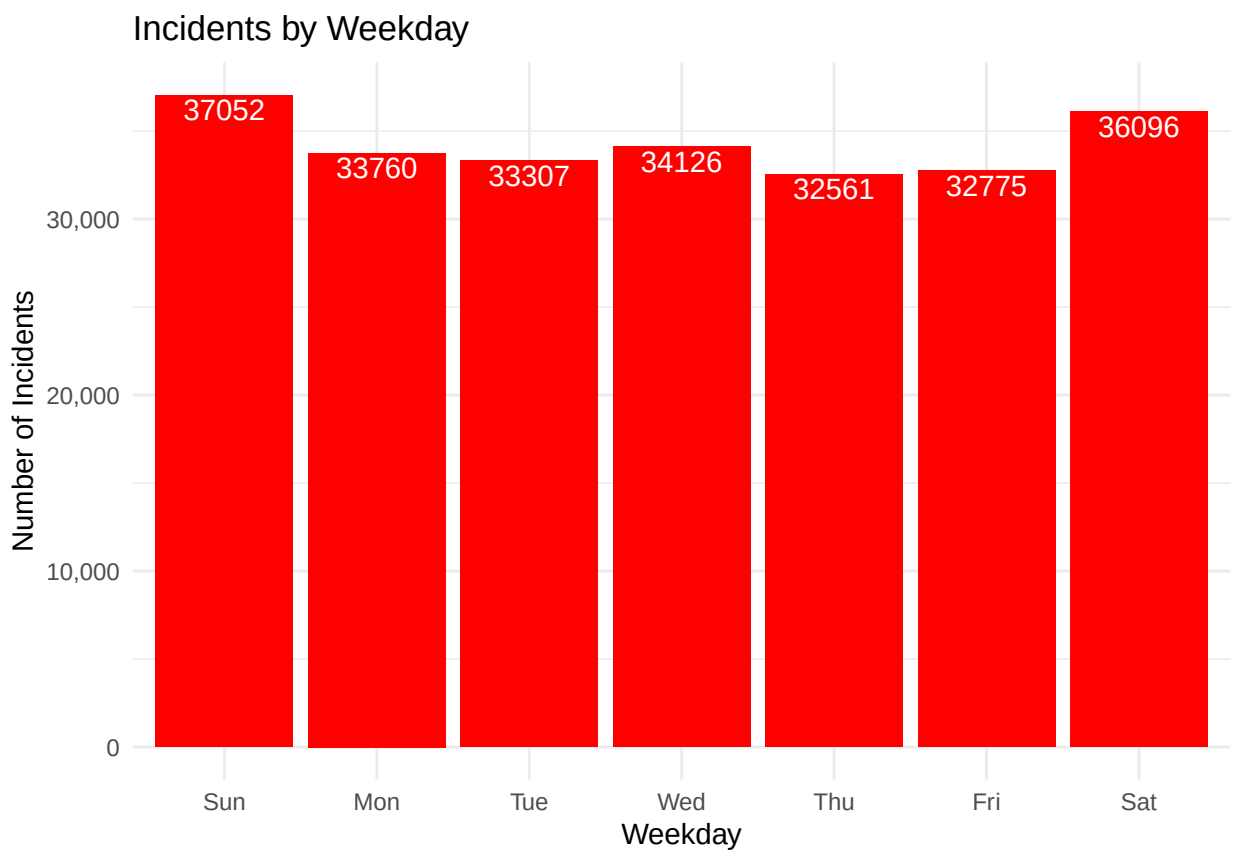


3.1.6 Insight:

Major holidays and the start of the year (e.g., `July 4th`, `New Year's` Day) see spikes in incidents, indicating possible connections between gun violence and certain significant dates.

```
# Extracting weekday and plotting incidents by weekday
gun_data <- gun_data %>% mutate(weekday = wday(date, label = TRUE))


gun_data %>%
```

```r
count(weekday) %>%
ggplot(aes(x = weekday, y = n)) +
geom_col(fill = 'red') +
scale_y_continuous(labels = comma) +
geom_text(aes(label = n), vjust = 1.2, color = "white", size = 4) +
labs(x = 'Weekday', y = 'Number of Incidents', title = 'Incidents by Weekday') +
theme_minimal()
```



3.1.7 Insight:

Incidents are more frequent on weekends, particularly `Sundays`, which may reflect increased social activity and gatherings.

```r
# Converting 'state' and 'city_or_county' columns to factors
gun_data <- gun_data %>%
  mutate(across(c(state, city_or_county), as.factor))

str(gun_data$state)
```

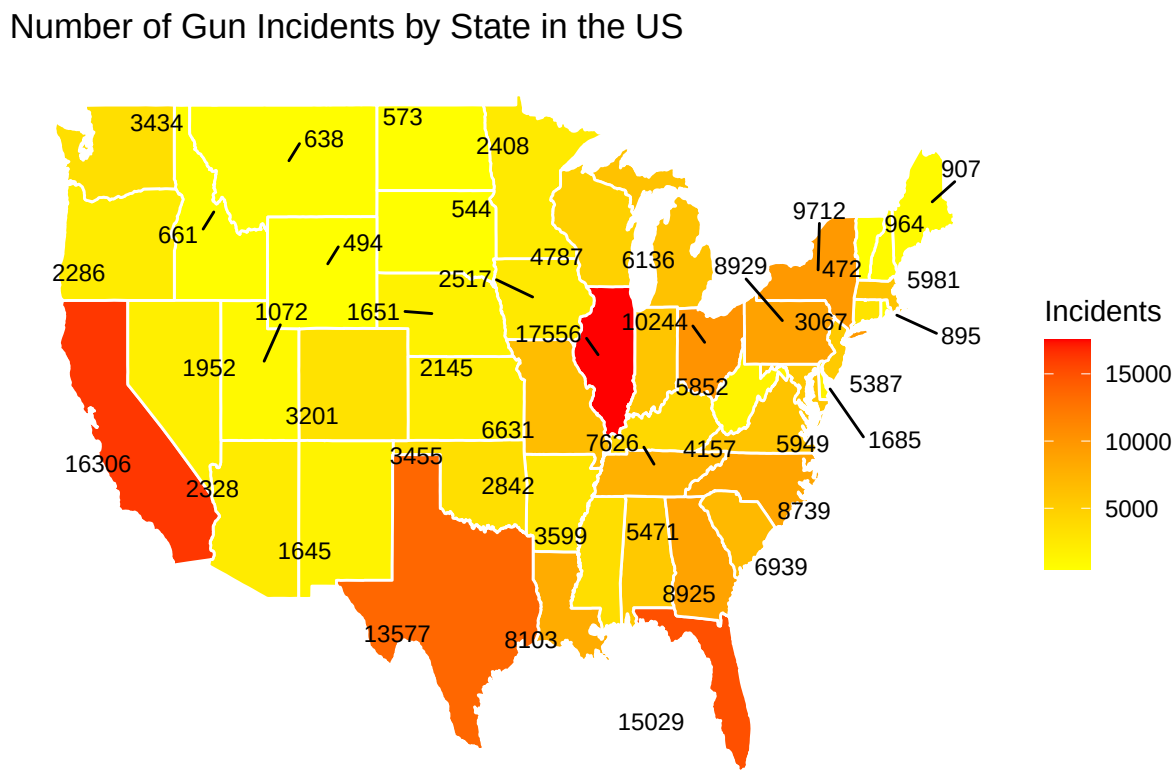
```
##  Factor w/ 51 levels "Alabama","Alaska",..: 39 5 36 6 34 37 32 19 5 21 ...
```

```r
str(gun_data$city_or_county)
```

```
##  Factor w/ 12898 levels "Abbeville","Abbotsford",..: 7178 4949 6607 451 4614 11730 89
```

```r
# Plotting incidents by state on the US map
us_states <- map_data("state")

gun_state_data <- gun_data %>%
  count(state) %>%
  mutate(state = tolower(state))

map_data <- us_states %>%
  left_join(gun_state_data, by = c("region" = "state"))

state_centroids <- map_data %>%
  group_by(region) %>%
  summarise(long = mean(range(long)), lat = mean(range(lat)), n = mean(n, na.rm = TRUE)
  filter(!is.na(n))

ggplot() +
  geom_polygon(data = map_data, aes(long, lat, group = group, fill = n), color = "white'
  geom_text_repel(data = state_centroids, aes(x = long, y = lat, label = round(n)),
```

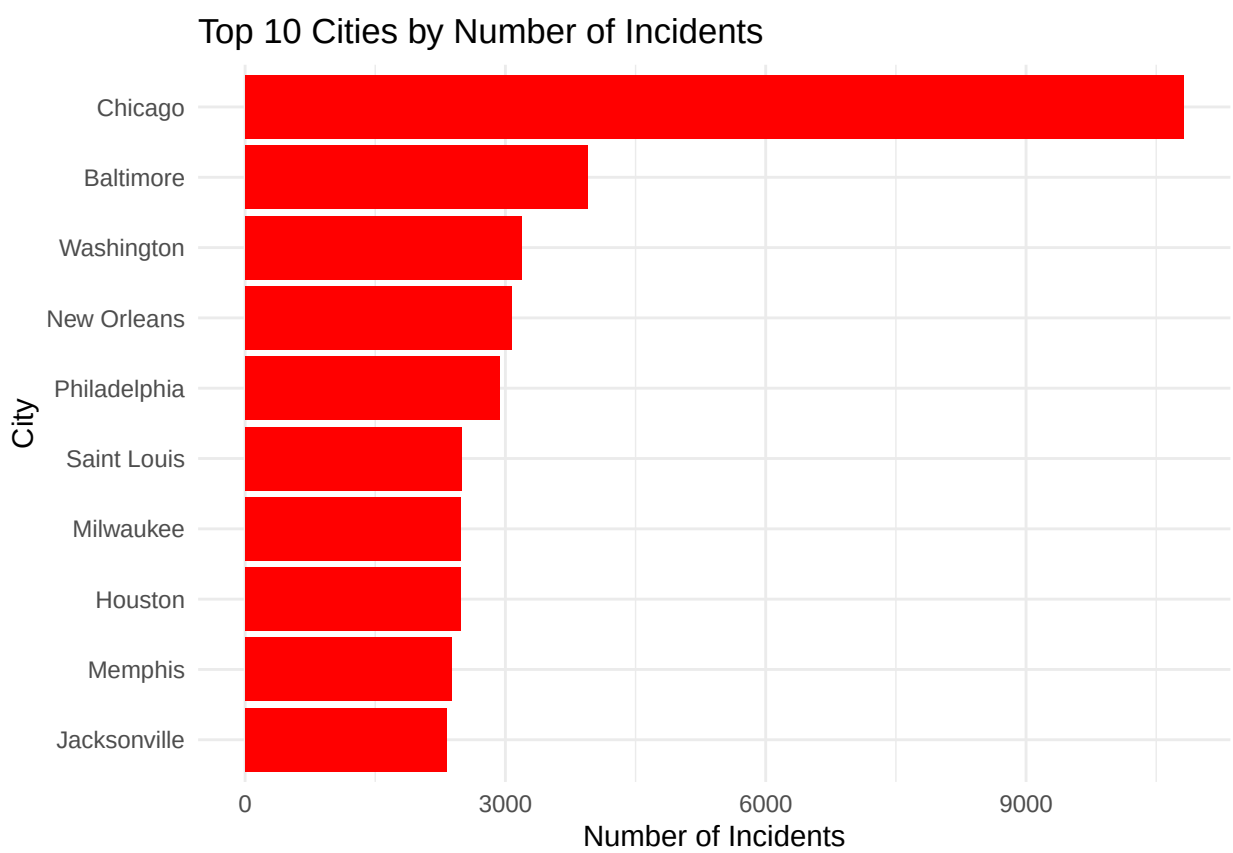
```
                 size = 3, color = "black", max.overlaps = 10, box.padding = 0.5, point
  scale_fill_gradient(low = "yellow", high = "red", na.value = "grey90", name = "Incider
  theme_minimal() +
  labs(title = "Number of Gun Incidents by State in the US", x = "", y = "") +
  theme(axis.text = element_blank(),
        axis.ticks = element_blank(),
        panel.grid = element_blank())
```



3.1.8 Insight:

States like `Illinois`, `California`, `Florida`, and `Texas` have the highest gun incident counts, highlighting regions that could benefit from targeted preventive measures.

```r
# Plotting the top 10 cities by the number of incidents
incidentsByCity <- gun_data %>%
  count(city = city_or_county, state, name = "cityIncidents")

incidentsByCity %>%
  slice_max(order_by = cityIncidents, n = 10) %>%
  ggplot(aes(x = reorder(city, cityIncidents), y = cityIncidents)) +
  geom_col(fill = 'red') +
  labs(x = 'City', y = 'Number of Incidents', title = 'Top 10 Cities by Number of Incide
  coord_flip() +
  theme_minimal()
```

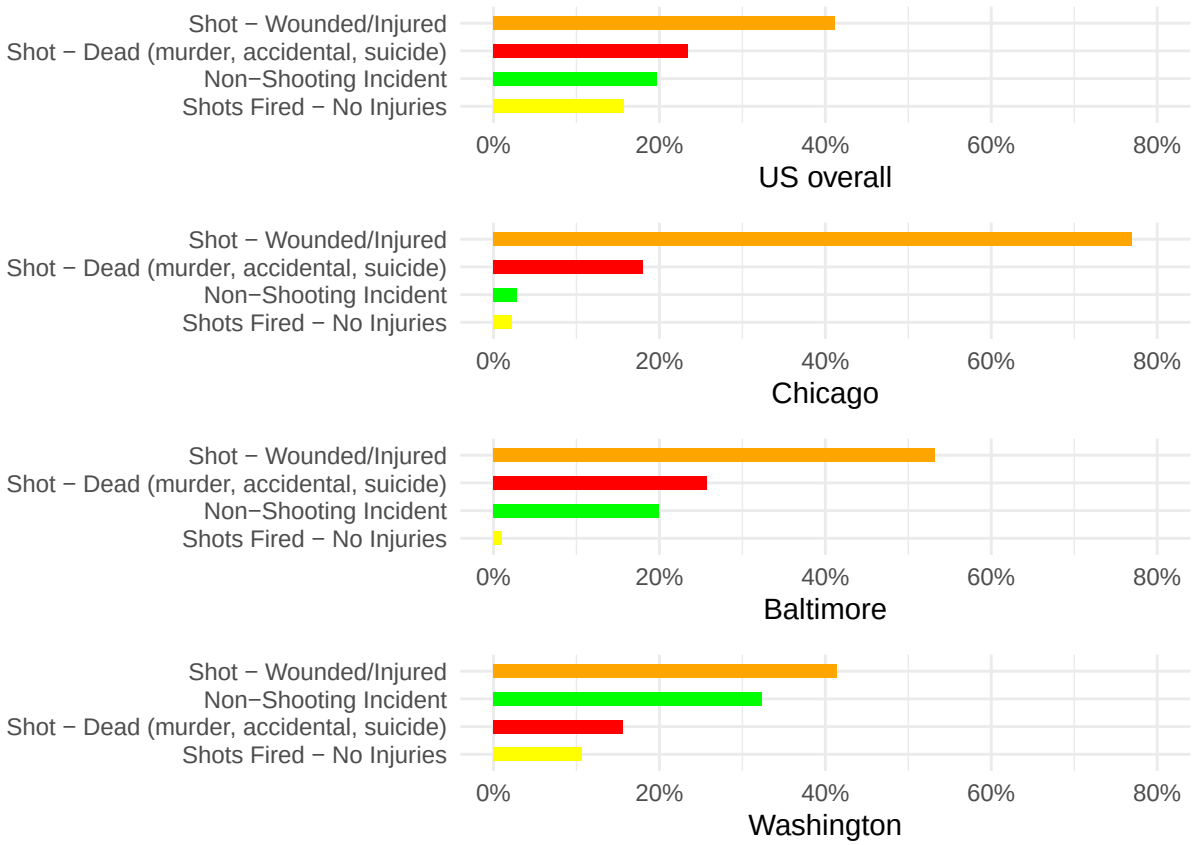


3.1.9 Insight:

`Chicago` leads in gun incidents, followed by cities like `Baltimore` and `Washington`, suggesting urban areas with high population densities are hotspots for gun violence.

```
# Defining the categories and color palette
overallCats <- c("Shot - Wounded/Injured", "Shot - Dead (murder, accidental, suicide)",
                 "Non-Shooting Incident", "Shots Fired - No Injuries")
coloursShot <- c("Shot - Wounded/Injured" = "orange",
                 "Shot - Dead (murder, accidental, suicide)" = "red",
                 "Non-Shooting Incident" = "green",
                 "Shots Fired - No Injuries" = "yellow")

# Data cleaning and processing function
prepare_data <- function(data) {
  data %>%
    mutate(incident_characteristics = gsub("\\|\\|", "|", incident_characteristics)) %>%
    separate_rows(incident_characteristics, sep = "\\|") %>%
    filter(incident_characteristics %in% overallCats)
}

# Plotting incidents for the entire US or a specific city
plot_incident_characteristics <- function(data, cityName = NULL, fixedX = 0.8) {
  plot_data <- if (is.null(cityName)) {
    data
  } else {
    filter(data, city_or_county == cityName)
  }

  plot_data %>%
    prepare_data() %>%
```

```r
    count(incident_characteristics) %>%
    ggplot(aes(x = reorder(incident_characteristics, n), y = n / sum(n), fill = incident
    geom_col(width = 0.5) +
    scale_fill_manual(values = coloursShot) +
    theme_minimal() +
    theme(legend.position = "none") +
    coord_flip() +
    scale_y_continuous(limits = c(0, fixedX), labels = percent) +
    labs(x = "", y = ifelse(is.null(cityName), "US overall", cityName))
}

# Loading the data
IncCharac <- prepare_data(gun_data)

# Creating plots for the US and specified cities
usOverallCats <- plot_incident_characteristics(IncCharac, fixedX = 0.8)
chicagoCats <- plot_incident_characteristics(IncCharac, 'Chicago', fixedX = 0.8)
baltimoreCats <- plot_incident_characteristics(IncCharac, 'Baltimore', fixedX = 0.8)
washingtonCats <- plot_incident_characteristics(IncCharac, 'Washington', fixedX = 0.8)

# plotting vertically
grid.arrange(usOverallCats, chicagoCats, baltimoreCats, washingtonCats, ncol = 1)
```



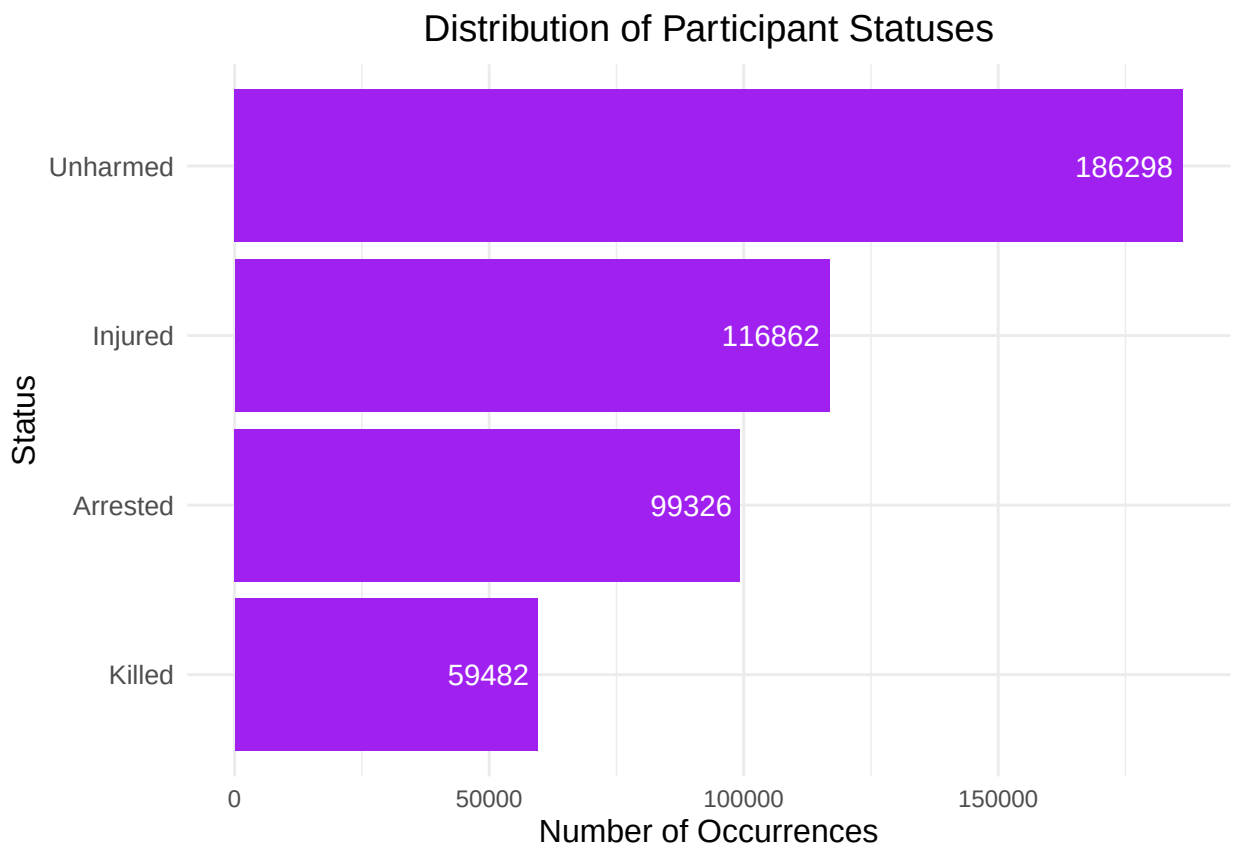
3.1.10 Insight:

Majority of incidents involve non-fatal injuries across the US and cities like `Chicago`, `Baltimore`, and `Washington`, while fatalities are less frequent but notable in these high-incident areas.

```
# Participant - Series
gun_data$participant_status[1:5]
```

```
## [1] "0::Arrested||1::Injured||2::Injured||3::Injured||4::Injured"
## [2] "0::Killed||1::Injured||2::Injured||3::Injured"
## [3] "0::Injured, Unharmed, Arrested||1::Unharmed, Arrested||2::Killed||3::Injured||4:
## [4] "0::Killed||1::Killed||2::Killed||3::Killed"
## [5] "0::Injured||1::Injured||2::Killed||3::Killed"
```

```r
# Splitting the 'participant_status' column into multiple rows
participant_status_cleaned <- gun_data %>%
  filter(participant_status != "") %>%
  mutate(participant_status = gsub("\\|\\|", "|", participant_status)) %>% # Replace "|
  select(incident_id, state, city_or_county, participant_status) %>%
  cSplit('participant_status', sep = '|', direction = "long") %>%
  mutate(status = sub("^[0-9]+::", "", participant_status)) %>%
  separate_rows(status, sep = ", ") %>%
  mutate(status = trimws(status)) %>%
  filter(!grepl("^[0-9]+:", status)) %>%
  drop_na(status)

# Counting the frequency of each participant status
status_distribution <- participant_status_cleaned %>%
  count(status) %>%
  arrange(desc(n))

# Plotting the participant status distribution
ggplot(status_distribution, aes(x = reorder(status, n), y = n)) +
  geom_col(fill = 'purple') +
  geom_text(aes(label = n), hjust = 1.1, color = "white", size = 4) +
  coord_flip() +
  labs(x = 'Status', y = 'Number of Occurrences', title = 'Distribution of Participant S
  theme_minimal() +
  theme(axis.text.y = element_text(size = 10),
        axis.title = element_text(size = 12),
        plot.title = element_text(hjust = 0.5, size = 14))
```

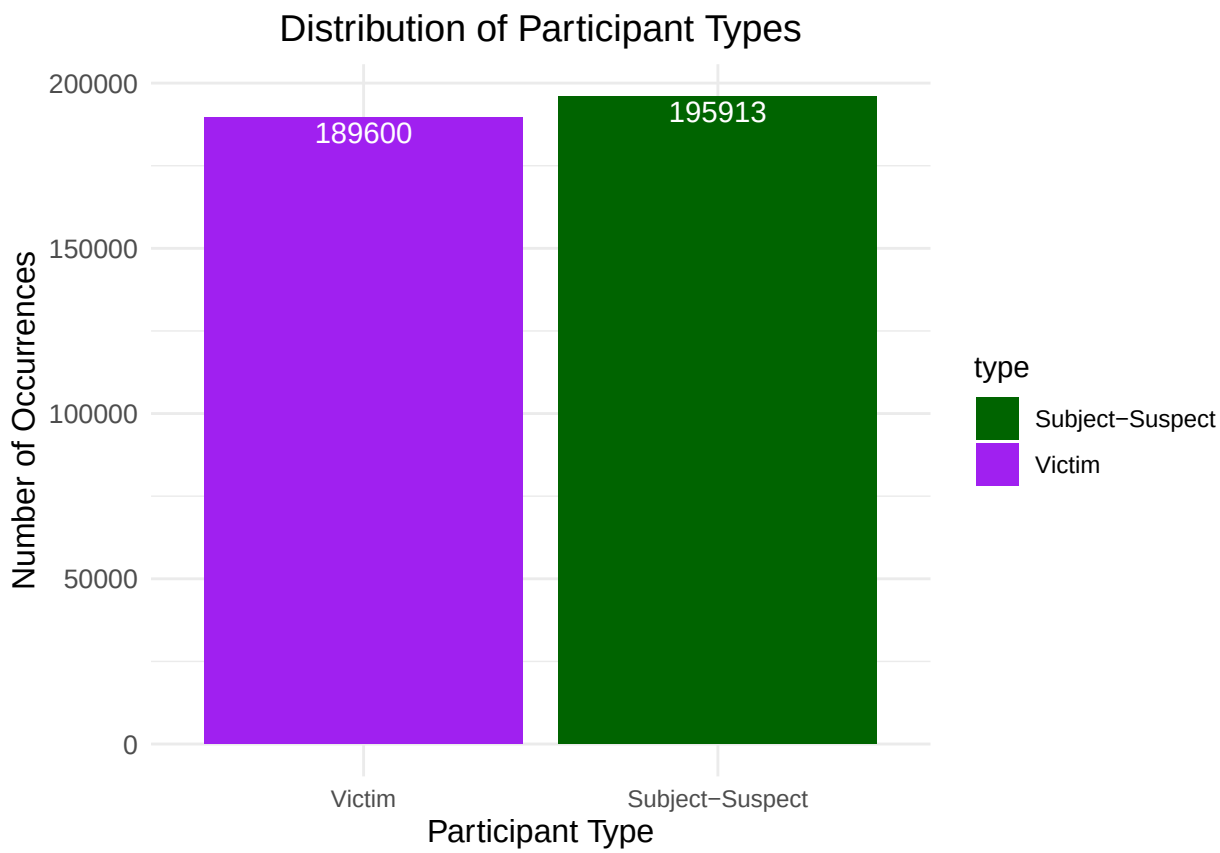



```
gun_data$participant_type[1:5]
```

```
## [1] "0::Victim||1::Victim||2::Victim||3::Victim||4::Subject-Suspect"
## [2] "0::Victim||1::Victim||2::Victim||3::Victim||4::Subject-Suspect"
## [3] "0::Subject-Suspect||1::Subject-Suspect||2::Victim||3::Victim||4::Victim"
## [4] "0::Victim||1::Victim||2::Victim||3::Subject-Suspect"
## [5] "0::Victim||1::Victim||2::Victim||3::Subject-Suspect"
```

```
# Splitting the 'participant_type' column into multiple rows
participant_type_cleaned <- gun_data %>%
  filter(participant_type != "") %>%
  mutate(participant_type = gsub("\\|\\|", "|", participant_type)) %>%
  select(incident_id, state, city_or_county, participant_type) %>%
  cSplit('participant_type', sep = '|', direction = "long") %>%
```

```r
  mutate(type = gsub("^[0-9]+::", "", participant_type)) %>%
  mutate(type = trimws(type)) %>%
  filter(type %in% c("Victim", "Subject-Suspect")) %>%
  drop_na(type)

# Counting the frequency of each participant type
type_distribution <- participant_type_cleaned %>%
  count(type) %>%
  arrange(desc(n))

# Plotting the distribution of participant types
ggplot(type_distribution, aes(x = reorder(type, n), y = n, fill = type)) +
  geom_col() +
  geom_text(aes(label = n), vjust = 1.2, color = "white", size = 4) +
  labs(x = 'Participant Type', y = 'Number of Occurrences', title = 'Distribution of Par
  theme_minimal() +
  theme(axis.text.y = element_text(size = 10),
        axis.title = element_text(size = 12),
        plot.title = element_text(hjust = 0.5, size = 14)) +
  scale_fill_manual(values = c("Subject-Suspect" = "darkgreen", "Victim" = "purple"))
```

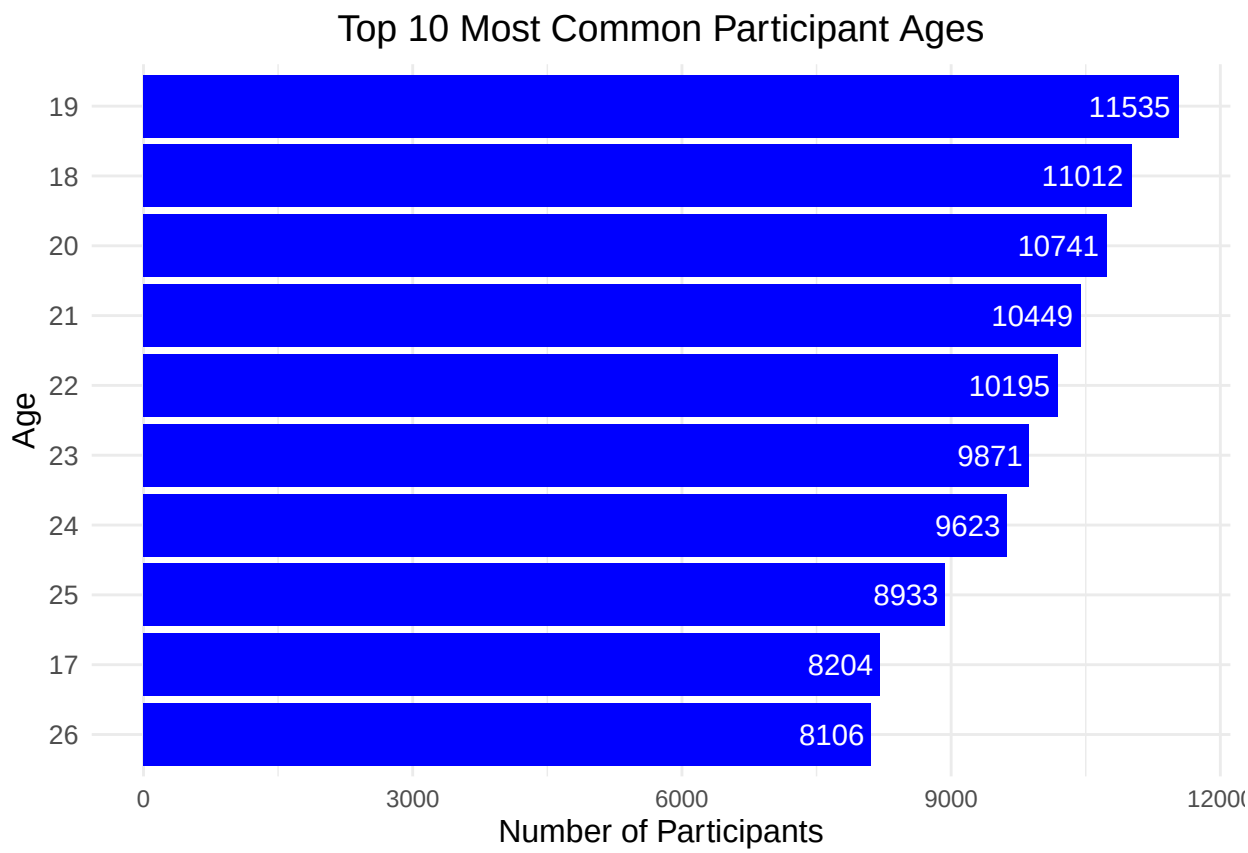



```r
gun_data$participant_age[1:5]
```

```
## [1] "0::20"                                "0::20"
## [3] "0::25||1::31||2::33||3::34||4::33" "0::29||1::33||2::56||3::33"
## [5] "0::18||1::46||2::14||3::47"
```

```r
# Splitting the 'participant_age' column into multiple rows
participant_age_cleaned <- gun_data %>%
  filter(participant_age != "") %>%
  mutate(participant_age = gsub("\\|\\|", "|", participant_age)) %>%
  select(incident_id, state, city_or_county, participant_age) %>%
  cSplit('participant_age', sep = '|', direction = "long") %>%
  mutate(age = as.numeric(sub(".*::", "", participant_age))) %>%
  drop_na(age)
```

```r
# Counting the top 10 most common ages
top10_age_distribution <- participant_age_cleaned %>%
  count(age) %>%
  arrange(desc(n)) %>%
  slice_max(order_by = n, n = 10)

# Plotting the top 10 most common participant ages
ggplot(top10_age_distribution, aes(x = reorder(age, n), y = n)) +
  geom_col(fill = 'blue') +
  geom_text(aes(label = n), hjust = 1.1, color = "white", size = 4) +
  coord_flip() +
  labs(x = 'Age', y = 'Number of Participants', title = 'Top 10 Most Common Participant
  theme_minimal() +
  theme(axis.text.y = element_text(size = 10),
        axis.title = element_text(size = 12),
        plot.title = element_text(hjust = 0.5, size = 14))
```

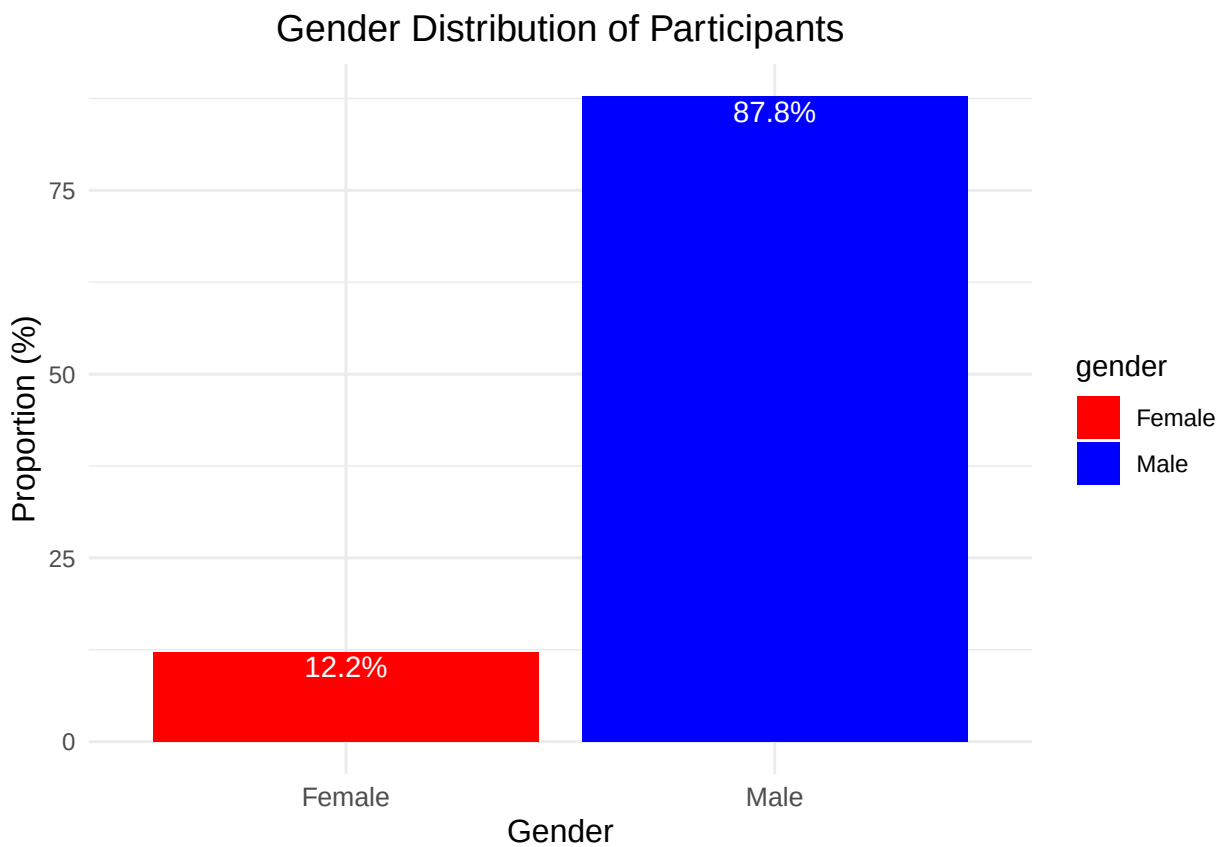



```
gun_data$participant_gender[1:5]
```

```
## [1] "0::Male||1::Male||3::Male||4::Female"
## [2] "0::Male"
## [3] "0::Male||1::Male||2::Male||3::Male||4::Male"
## [4] "0::Female||1::Male||2::Male||3::Male"
## [5] "0::Female||1::Male||2::Male||3::Female"
```

```
# Splitting the 'participant_gender' column into multiple rows
participant_gender_cleaned <- gun_data %>%
  filter(participant_gender != "") %>%
  mutate(participant_gender = gsub("\\|\\|", "|", participant_gender)) %>%
  select(incident_id, state, city_or_county, participant_gender) %>%
  cSplit('participant_gender', sep = '|', direction = "long") %>%
```

```r
  mutate(gender = sub(".*::", "", participant_gender)) %>%
  drop_na(gender)

# Grouping all "Male" and "Female" together
gender_aggregated <- participant_gender_cleaned %>%
  filter(gender %in% c("Male", "Female")) %>%
  count(gender) %>%
  mutate(proportion = n / sum(n) * 100)

# Plotting the gender distribution
ggplot(gender_aggregated, aes(x = gender, y = proportion, fill = gender)) +
  geom_col() +
  geom_text(aes(label = paste0(round(proportion, 1), "%")),
            vjust = 1.2, color = "white", size = 4) +
  labs(x = 'Gender', y = 'Proportion (%)', title = 'Gender Distribution of Participants'
  theme_minimal() +
  theme(axis.text.x = element_text(size = 10),
        axis.title = element_text(size = 12),
        plot.title = element_text(hjust = 0.5, size = 14)) +
  scale_fill_manual(values = c("Male" = "blue", "Female" = "red"))
```




```
gun_data$incident_characteristics[1:5]
```

```
## [1] "Shot - Wounded/Injured||Mass Shooting (4+ victims injured or killed excluding th
## [2] "Shot - Wounded/Injured||Shot - Dead (murder, accidental, suicide)||Mass Shooting
## [3] "Shot - Wounded/Injured||Shot - Dead (murder, accidental, suicide)||Shots Fired -
## [4] "Shot - Dead (murder, accidental, suicide)||Officer Involved Incident||Officer In
## [5] "Shot - Wounded/Injured||Shot - Dead (murder, accidental, suicide)||Suicide^||Mur
```

```
# Splitting the 'incident_characteristics' column into multiple rows
incident_characteristics_cleaned <- gun_data %>%
  filter(incident_characteristics != "") %>% # Filter out empty values
  mutate(incident_characteristics = gsub("\\|\\|", "|", incident_characteristics)) %>%
  select(incident_id, state, city_or_county, incident_characteristics) %>%
  cSplit('incident_characteristics', sep = '|', direction = "long") %>%
```

```r
  mutate(characteristic = trimws(incident_characteristics)) %>%
  drop_na(characteristic)

# Counting the frequency of each characteristic
top10_characteristics <- incident_characteristics_cleaned %>%
  count(characteristic) %>%
  arrange(desc(n)) %>%
  slice_max(order_by = n, n = 10)

# Plotting the top 10 most common incident characteristics
ggplot(top10_characteristics, aes(x = reorder(characteristic, n), y = n)) +
  geom_col(fill = 'blue') +
  coord_flip() +
  labs(x = 'Incident Characteristic', y = 'Number of Incidents', title = 'Top 10 Most Co
  theme_minimal() +
  theme(axis.text.y = element_text(size = 10),
        axis.title = element_text(size = 12),
        plot.title = element_text(hjust = 0.5, size = 14))
```

3.1.11 Insights:

1. **Participant Status**: Most participants were unharmed, with substantial numbers also injured, arrested, or killed.
2. **Participant Type**: Similar numbers of victims and suspects, highlighting the balanced representation of both groups.
3. **Participant Age**: Ages 18-25 are the most involved, showing teenager's are more commonly linked to these incidents.
4. **Participant Gender**: Males overwhelmingly dominate incidents.
5. **Incident Characteristics**: Non-fatal shootings are most common, followed by fatal incidents and non-shooting incidents, showing a high prevalence of injuries over deaths.

3.2 Code for Wordcloud

```r
library(tidyverse)
library(tidytext)
library(tm)
library(wordcloud)
library(RColorBrewer)

# Extracting and preprocessing the 'location_description' and 'n_killed' columns
location_data <- gun_data %>%
  select(location = location_description, killed = n_killed) %>%
  filter(!is.na(location), !is.na(killed)) %>%
  mutate(location = tolower(location),
         location = str_replace_all(location, "[[:punct:]]", " "),
         location = str_squish(location))

# Tokenizing the text and removing stopwords
location_words <- location_data %>%
  unnest_tokens(word, location, drop = FALSE) %>%
  anti_join(stop_words)

# Calculating word frequencies
word_freqs <- location_words %>%
  count(word, sort = TRUE)
head(word_freqs)
```

```
##           word    n
## 1   apartments 4271
## 2         park 3364
## 3       school 1835
```

```
## 4 neighborhood 1408
## 5      airport 1070
## 6          inn 1007
```

```r
# Plotting the basic word cloud
set.seed(123)
wordcloud(words = word_freqs$word, freq = word_freqs$n,
          min.freq = 10, max.words = 200, random.order = FALSE,
          colors = brewer.pal(8, "Dark2"))
```

```r
# Calculating the total number of deaths for each location
location_weights <- location_data %>%
  group_by(location) %>%
  summarise(total_killed = sum(killed))
```

```r
# Associating words with the total deaths at their locations
weighted_words <- location_words %>%
  inner_join(location_weights, by = "location") %>%
  group_by(word) %>%
  summarise(freq = n(),
            weighted_freq = sum(total_killed)) %>%
  arrange(desc(weighted_freq))
head(weighted_words)
```

```
## # A tibble: 6 x 3
##   word     freq weighted_freq
##   <chr>   <int>         <int>
## 1 walmart   388         26385
## 2 austin    429         15558
## 3 7         260         10361
## 4 eleven    215         10309
## 5 park     3364          9309
## 6 motel     458          7916
```

```r
# Plotting the weighted word cloud
set.seed(123)
wordcloud(words = weighted_words$word, freq = weighted_words$weighted_freq,
          min.freq = 10, max.words = 200, random.order = FALSE,
          colors = brewer.pal(8, "Dark2"))
```

3.2.1 Insights:

1. **Top Locations for Incidents**: Common locations for gun violence incidents include `apartments`, `parks`, and `schools`, indicating frequent occurrences in public and residential areas.

2. **Locations with High Fatalities**: `Walmart`, `Austin`, and `7-Eleven` locations are associated with higher death counts, suggesting these may be high-risk areas.

These insights point to specific environments that could benefit from targeted safety measures.

3.3 Code for RandomForest

```
library(caret)
library(fastDummies)
library(randomForest)

# Dropping missing values
gun_data_clean <- gun_data %>% drop_na()

# Convert 'state', 'month', and 'weekday' to factor type
gun_data_clean$state <- as.factor(gun_data_clean$state)
gun_data_clean$month <- as.factor(gun_data_clean$month)
gun_data_clean$weekday <- as.factor(gun_data_clean$weekday)

# Applying one-hot encoding using fastDummies
gun_data_encoded <- dummy_cols(
  gun_data_clean,
  select_columns = c("state", "month", "weekday"),
  remove_first_dummy = FALSE,
  remove_selected_columns = TRUE
)

# Creating a new variable for total casualties
gun_data_encoded$total_casualties <- gun_data_encoded$n_killed + gun_data_encoded$n_inju

# Creating the 'high_casualty' variable
gun_data_encoded$high_casualty <- ifelse(gun_data_encoded$total_casualties >= 3, 1, 0)

# List of columns to remove
columns_to_remove <- c(
```

```r
  "incident_id", "date", "city_or_county", "address", "incident_url", "source_url",
  "incident_url_fields_missing", "gun_stolen", "gun_type", "incident_characteristics",
  "latitude", "location_description", "longitude", "notes", "participant_age",
  "participant_age_group", "participant_gender", "participant_name",
  "participant_relationship", "participant_status", "participant_type",
  "sources", "day", "date2", "n_injured", "n_killed")

# Removing the specified columns
gun_data_encoded <- gun_data_encoded[, !(names(gun_data_encoded) %in% columns_to_remove)

# Splitting the dataset into training and testing sets
set.seed(123)
train_index <- sample(seq_len(nrow(gun_data_encoded)), size = 0.7 * nrow(gun_data_encode
train_data <- gun_data_encoded[train_index, ]
test_data <- gun_data_encoded[-train_index, ]

# Ensuring column names have no spaces
names(train_data) <- gsub(" ", "_", names(train_data))
names(test_data) <- gsub(" ", "_", names(test_data))

# Identifying numeric columns for scaling, EXCLUDING 'total_casualties' and 'high_casu
numeric_columns <- sapply(train_data, is.numeric)
numeric_columns <- names(train_data)[numeric_columns]
numeric_columns <- setdiff(numeric_columns, c("total_casualties", "high_casualty"))

# Scaling numeric columns
scaling_params <- preProcess(train_data[numeric_columns], method = c("center", "scale"))
train_data_scaled <- train_data
train_data_scaled[numeric_columns] <- predict(scaling_params, train_data[numeric_columns
test_data_scaled <- test_data
```

```r
test_data_scaled[numeric_columns] <- predict(scaling_params, test_data[numeric_columns])

# Removing the 'state_Vermont' column if present
if ("state_Vermont" %in% names(train_data_scaled)) {
  train_data_scaled <- train_data_scaled[, !(names(train_data_scaled) %in% "state_Vermon
}
if ("state_Vermont" %in% names(test_data_scaled)) {
  test_data_scaled <- test_data_scaled[, !(names(test_data_scaled) %in% "state_Vermont")
}

# Convert 'high_casualty' to a factor for classification
train_data_scaled$high_casualty <- as.factor(train_data_scaled$high_casualty)
test_data_scaled$high_casualty <- as.factor(test_data_scaled$high_casualty)
summary(train_data_scaled)
```

```
##  congressional_district n_guns_involved    state_house_district
##  Min.   :-0.9689        Min.   :-0.07545   Min.   :-1.2224
##  1st Qu.:-0.7328        1st Qu.:-0.07545   1st Qu.:-0.7698
##  Median :-0.3787        Median :-0.07545   Median :-0.2042
##  Mean   : 0.0000        Mean   : 0.00000   Mean   : 0.0000
##  3rd Qu.: 0.3296        3rd Qu.:-0.07545   3rd Qu.: 0.6330
##  Max.   : 5.2875        Max.   :79.90351   Max.   :17.1730
##  state_senate_district      year            quarter        state_Alabama
##  Min.   :-1.3628       Min.   :-3.3780   Min.   :-1.2757   Min.   :-0.1522
##  1st Qu.:-0.8708       1st Qu.:-0.3823   1st Qu.:-1.2757   1st Qu.:-0.1522
##  Median :-0.1678       Median : 0.6163   Median : 0.4459   Median :-0.1522
##  Mean   : 0.0000       Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.: 0.6757       3rd Qu.: 0.6163   3rd Qu.: 1.3066   3rd Qu.:-0.1522
##  Max.   : 2.9954       Max.   : 1.6149   Max.   : 1.3066   Max.   : 6.5685
##   state_Alaska      state_Arizona     state_Arkansas    state_California
```

```
##  Min.   :-0.02336   Min.   :-0.1095   Min.   :-0.1186   Min.   :-0.2955
##  1st Qu.:-0.02336   1st Qu.:-0.1095   1st Qu.:-0.1186   1st Qu.:-0.2955
##  Median :-0.02336   Median :-0.1095   Median :-0.1186   Median :-0.2955
##  Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.:-0.02336   3rd Qu.:-0.1095   3rd Qu.:-0.1186   3rd Qu.:-0.2955
##  Max.   :42.80866   Max.   : 9.1283   Max.   : 8.4303   Max.   : 3.3843
##  state_Colorado    state_Connecticut state_Delaware
##  Min.   :-0.1267   Min.   :-0.1256   Min.   :-0.08698
##  1st Qu.:-0.1267   1st Qu.:-0.1256   1st Qu.:-0.08698
##  Median :-0.1267   Median :-0.1256   Median :-0.08698
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.00000
##  3rd Qu.:-0.1267   3rd Qu.:-0.1256   3rd Qu.:-0.08698
##  Max.   : 7.8948   Max.   : 7.9634   Max.   :11.49706
##  state_District_of_Columbia state_Florida     state_Georgia
##  Min.   : -0.00504          Min.   :-0.2855   Min.   :-0.2039
##  1st Qu.: -0.00504          1st Qu.:-0.2855   1st Qu.:-0.2039
##  Median : -0.00504          Median :-0.2855   Median :-0.2039
##  Mean   :  0.00000          Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.: -0.00504          3rd Qu.:-0.2855   3rd Qu.:-0.2039
##  Max.   :198.54722          Max.   : 3.5027   Max.   : 4.9036
##   state_Hawaii       state_Idaho       state_Illinois    state_Indiana
##  Min.   :-0.03953   Min.   :-0.06404   Min.   :-0.3215   Min.   :-0.1813
##  1st Qu.:-0.03953   1st Qu.:-0.06404   1st Qu.:-0.3215   1st Qu.:-0.1813
##  Median :-0.03953   Median :-0.06404   Median :-0.3215   Median :-0.1813
##  Mean   : 0.00000   Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.:-0.03953   3rd Qu.:-0.06404   3rd Qu.:-0.3215   3rd Qu.:-0.1813
##  Max.   :25.29838   Max.   :15.61593   Max.   : 3.1104   Max.   : 5.5147
##    state_Iowa       state_Kansas     state_Kentucky     state_Louisiana
##  Min.   :-0.1081   Min.   :-0.1021   Min.   :-0.06424   Min.   :-0.1895
##  1st Qu.:-0.1081   1st Qu.:-0.1021   1st Qu.:-0.06424   1st Qu.:-0.1895
```

```
##  Median :-0.1081   Median :-0.1021   Median :-0.06424   Median :-0.1895
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.00000   Mean   : 0.0000
##  3rd Qu.:-0.1081   3rd Qu.:-0.1021   3rd Qu.:-0.06424   3rd Qu.:-0.1895
##  Max.   : 9.2543   Max.   : 9.7969   Max.   :15.56746   Max.   : 5.2782
##   state_Maine       state_Maryland    state_Massachusetts state_Michigan
##  Min.   :-0.06941   Min.   :-0.1406   Min.   :-0.08735    Min.   :-0.1834
##  1st Qu.:-0.06941   1st Qu.:-0.1406   1st Qu.:-0.08735    1st Qu.:-0.1834
##  Median :-0.06941   Median :-0.1406   Median :-0.08735    Median :-0.1834
##  Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.00000    Mean   : 0.0000
##  3rd Qu.:-0.06941   3rd Qu.:-0.1406   3rd Qu.:-0.08735    3rd Qu.:-0.1834
##  Max.   :14.40771   Max.   : 7.1110   Max.   :11.44845    Max.   : 5.4522
##  state_Minnesota     state_Mississippi state_Missouri    state_Montana
##  Min.   : -0.00504   Min.   :-0.1241   Min.   :-0.1848   Min.   :-0.02666
##  1st Qu.: -0.00504   1st Qu.:-0.1241   1st Qu.:-0.1848   1st Qu.:-0.02666
##  Median : -0.00504   Median :-0.1241   Median :-0.1848   Median :-0.02666
##  Mean   :  0.00000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.00000
##  3rd Qu.: -0.00504   3rd Qu.:-0.1241   3rd Qu.:-0.1848   3rd Qu.:-0.02666
##  Max.   :198.54722   Max.   : 8.0575   Max.   : 5.4098   Max.   :37.50905
##  state_Nebraska       state_Nevada      state_New_Hampshire state_New_Jersey
##  Min.   : -0.00356   Min.   :-0.09188   Min.   :-0.03227    Min.   :-0.1562
##  1st Qu.: -0.00356   1st Qu.:-0.09188   1st Qu.:-0.03227    1st Qu.:-0.1562
##  Median : -0.00356   Median :-0.09188   Median :-0.03227    Median :-0.1562
##  Mean   :  0.00000   Mean   : 0.00000   Mean   : 0.00000    Mean   : 0.0000
##  3rd Qu.: -0.00356   3rd Qu.:-0.09188   3rd Qu.:-0.03227    3rd Qu.:-0.1562
##  Max.   :280.78996   Max.   :10.88396   Max.   :30.99213    Max.   : 6.4000
##  state_New_Mexico   state_New_York   state_North_Carolina state_North_Dakota
##  Min.   :-0.09382   Min.   :-0.219   Min.   :-0.1881      Min.   :-0.0556
##  1st Qu.:-0.09382   1st Qu.:-0.219   1st Qu.:-0.1881      1st Qu.:-0.0556
##  Median :-0.09382   Median :-0.219   Median :-0.1881      Median :-0.0556
##  Mean   : 0.00000   Mean   : 0.000   Mean   : 0.0000      Mean   : 0.0000
```

```
##  3rd Qu.:-0.09382   3rd Qu.:-0.219   3rd Qu.:-0.1881      3rd Qu.:-0.0556
##  Max.   :10.65828   Max.   : 4.565   Max.   : 5.3156      Max.   :17.9850
##    state_Ohio      state_Oklahoma     state_Oregon     state_Pennsylvania
##  Min.   :-0.2419   Min.   :-0.1288   Min.   :-0.1122   Min.   :-0.0995
##  1st Qu.:-0.2419   1st Qu.:-0.1288   1st Qu.:-0.1122   1st Qu.:-0.0995
##  Median :-0.2419   Median :-0.1288   Median :-0.1122   Median :-0.0995
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.:-0.2419   3rd Qu.:-0.1288   3rd Qu.:-0.1122   3rd Qu.:-0.0995
##  Max.   : 4.1347   Max.   : 7.7659   Max.   : 8.9136   Max.   :10.0497
##  state_Rhode_Island state_South_Carolina state_South_Dakota state_Tennessee
##  Min.   :-0.06522   Min.   :-0.1752      Min.   :-0.0556    Min.   :-0.1861
##  1st Qu.:-0.06522   1st Qu.:-0.1752      1st Qu.:-0.0556    1st Qu.:-0.1861
##  Median :-0.06522   Median :-0.1752      Median :-0.0556    Median :-0.1861
##  Mean   : 0.00000   Mean   : 0.0000      Mean   : 0.0000    Mean   : 0.0000
##  3rd Qu.:-0.06522   3rd Qu.:-0.1752      3rd Qu.:-0.0556    3rd Qu.:-0.1861
##  Max.   :15.33166   Max.   : 5.7066      Max.   :17.9850    Max.   : 5.3737
##   state_Texas        state_Utah       state_Virginia    state_Washington
##  Min.   :-0.2094   Min.   :-0.07785   Min.   :-0.1551   Min.   :-0.1384
##  1st Qu.:-0.2094   1st Qu.:-0.07785   1st Qu.:-0.1551   1st Qu.:-0.1384
##  Median :-0.2094   Median :-0.07785   Median :-0.1551   Median :-0.1384
##  Mean   : 0.0000   Mean   : 0.00000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.:-0.2094   3rd Qu.:-0.07785   3rd Qu.:-0.1551   3rd Qu.:-0.1384
##  Max.   : 4.7748   Max.   :12.84474   Max.   : 6.4459   Max.   : 7.2275
##  state_West_Virginia state_Wisconsin   state_Wyoming        month_Jan
##  Min.   :-0.0872     Min.   :-0.1516   Min.   :-0.04322   Min.   :-0.3428
##  1st Qu.:-0.0872     1st Qu.:-0.1516   1st Qu.:-0.04322   1st Qu.:-0.3428
##  Median :-0.0872     Median :-0.1516   Median :-0.04322   Median :-0.3428
##  Mean   : 0.0000     Mean   : 0.0000   Mean   : 0.00000   Mean   : 0.0000
##  3rd Qu.:-0.0872     3rd Qu.:-0.1516   3rd Qu.:-0.04322   3rd Qu.:-0.3428
##  Max.   :11.4678     Max.   : 6.5969   Max.   :23.13771   Max.   : 2.9173
```

```
##    month_Feb         month_Mar         month_Apr         month_May
##  Min.   :-0.3023   Min.   :-0.3282   Min.   :-0.2733   Min.   :-0.2797
##  1st Qu.:-0.3023   1st Qu.:-0.3282   1st Qu.:-0.2733   1st Qu.:-0.2797
##  Median :-0.3023   Median :-0.3282   Median :-0.2733   Median :-0.2797
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.:-0.3023   3rd Qu.:-0.3282   3rd Qu.:-0.2733   3rd Qu.:-0.2797
##  Max.   : 3.3077   Max.   : 3.0465   Max.   : 3.6585   Max.   : 3.5755
##    month_Jun         month_Jul         month_Aug         month_Sep
##  Min.   :-0.2689   Min.   :-0.2805   Min.   :-0.3107   Min.   :-0.2939
##  1st Qu.:-0.2689   1st Qu.:-0.2805   1st Qu.:-0.3107   1st Qu.:-0.2939
##  Median :-0.2689   Median :-0.2805   Median :-0.3107   Median :-0.2939
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.:-0.2689   3rd Qu.:-0.2805   3rd Qu.:-0.3107   3rd Qu.:-0.2939
##  Max.   : 3.7183   Max.   : 3.5654   Max.   : 3.2183   Max.   : 3.4022
##    month_Oct         month_Nov         month_Dec        weekday_Sun
##  Min.   :-0.3074   Min.   :-0.3001   Min.   :-0.3237   Min.   :-0.4219
##  1st Qu.:-0.3074   1st Qu.:-0.3001   1st Qu.:-0.3237   1st Qu.:-0.4219
##  Median :-0.3074   Median :-0.3001   Median :-0.3237   Median :-0.4219
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.:-0.3074   3rd Qu.:-0.3001   3rd Qu.:-0.3237   3rd Qu.:-0.4219
##  Max.   : 3.2531   Max.   : 3.3317   Max.   : 3.0890   Max.   : 2.3704
##   weekday_Mon       weekday_Tue       weekday_Wed       weekday_Thu
##  Min.   :-0.4058   Min.   :-0.4067   Min.   :-0.4114   Min.   :-0.4009
##  1st Qu.:-0.4058   1st Qu.:-0.4067   1st Qu.:-0.4114   1st Qu.:-0.4009
##  Median :-0.4058   Median :-0.4067   Median :-0.4114   Median :-0.4009
##  Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000   Mean   : 0.0000
##  3rd Qu.:-0.4058   3rd Qu.:-0.4067   3rd Qu.:-0.4114   3rd Qu.:-0.4009
##  Max.   : 2.4640   Max.   : 2.4587   Max.   : 2.4309   Max.   : 2.4946
##   weekday_Fri       weekday_Sat      total_casualties  high_casualty
##  Min.   :-0.3983   Min.   :-0.4126   Min.   :  0.000   0:76871
```

```
##  1st Qu.:-0.3983   1st Qu.:-0.4126   1st Qu.:  0.000   1: 1974
##  Median :-0.3983   Median :-0.4126   Median :  1.000
##  Mean   : 0.0000   Mean   : 0.0000   Mean   :  0.712
##  3rd Qu.:-0.3983   3rd Qu.:-0.4126   3rd Qu.:  1.000
##  Max.   : 2.5109   Max.   : 2.4235   Max.   :103.000
```

```r
summary(test_data_scaled)
```

```
##  congressional_district n_guns_involved    state_house_district
##  Min.   :-0.96891       Min.   :-0.07545   Min.   :-1.222350
##  1st Qu.:-0.73282       1st Qu.:-0.07545   1st Qu.:-0.769820
##  Median :-0.37868       Median :-0.07545   Median :-0.204158
##  Mean   :-0.01313       Mean   : 0.00364   Mean   :-0.007419
##  3rd Qu.: 0.32959       3rd Qu.:-0.07545   3rd Qu.: 0.633023
##  Max.   : 5.28749       Max.   :79.70306   Max.   :19.141490
##  state_senate_district      year             quarter
##  Min.   :-1.362831     Min.   :-3.37796   Min.   :-1.275664
##  1st Qu.:-0.870774     1st Qu.:-0.38225   1st Qu.:-1.275664
##  Median :-0.167834     Median : 0.61631   Median : 0.445869
##  Mean   :-0.000761     Mean   : 0.00475   Mean   : 0.003119
##  3rd Qu.: 0.675693     3rd Qu.: 0.61631   3rd Qu.: 1.306636
##  Max.   : 5.174505     Max.   : 1.61488   Max.   : 1.306636
##  state_Alabama        state_Alaska      state_Arizona       state_Arkansas
##  Min.   :-0.152239   Min.   :-0.02336   Min.   :-0.109549   Min.   :-0.118619
##  1st Qu.:-0.152239   1st Qu.:-0.02336   1st Qu.:-0.109549   1st Qu.:-0.118619
##  Median :-0.152239   Median :-0.02336   Median :-0.109549   Median :-0.118619
##  Mean   :-0.002871   Mean   : 0.00960   Mean   : 0.007185   Mean   :-0.000724
##  3rd Qu.:-0.152239   3rd Qu.:-0.02336   3rd Qu.:-0.109549   3rd Qu.:-0.118619
##  Max.   : 6.568530   Max.   :42.80866   Max.   : 9.128261   Max.   : 8.430272
##  state_California    state_Colorado      state_Connecticut
```

```
##  Min.   :-0.295476   Min.   :-0.126663   Min.   :-0.125574
##  1st Qu.:-0.295476   1st Qu.:-0.126663   1st Qu.:-0.125574
##  Median :-0.295476   Median :-0.126663   Median :-0.125574
##  Mean   : 0.003016   Mean   : 0.001762   Mean   :-0.008756
##  3rd Qu.:-0.295476   3rd Qu.:-0.126663   3rd Qu.:-0.125574
##  Max.   : 3.384325   Max.   : 7.894842   Max.   : 7.963356
##  state_Delaware      state_District_of_Columbia state_Florida
##  Min.   :-0.086978   Min.   : -0.00504          Min.   :-0.285490
##  1st Qu.:-0.086978   1st Qu.: -0.00504          1st Qu.:-0.285490
##  Median :-0.086978   Median : -0.00504          Median :-0.285490
##  Mean   :-0.005731   Mean   :  0.01259          Mean   :-0.002197
##  3rd Qu.:-0.086978   3rd Qu.: -0.00504          3rd Qu.:-0.285490
##  Max.   :11.497062   Max.   :198.54722          Max.   : 3.502705
##  state_Georgia        state_Hawaii        state_Idaho
##  Min.   :-0.203927   Min.   :-0.039528   Min.   :-0.064036
##  1st Qu.:-0.203927   1st Qu.:-0.039528   1st Qu.:-0.064036
##  Median :-0.203927   Median :-0.039528   Median :-0.064036
##  Mean   : 0.001186   Mean   : 0.005463   Mean   : 0.001392
##  3rd Qu.:-0.203927   3rd Qu.:-0.039528   3rd Qu.:-0.064036
##  Max.   : 4.903647   Max.   :25.298382   Max.   :15.615926
##  state_Illinois       state_Indiana         state_Iowa
##  Min.   :-0.3214944   Min.   :-0.181331   Min.   :-0.108057
##  1st Qu.:-0.3214944   1st Qu.:-0.181331   1st Qu.:-0.108057
##  Median :-0.3214944   Median :-0.181331   Median :-0.108057
##  Mean   :-0.0006556   Mean   : 0.004935   Mean   :-0.001663
##  3rd Qu.:-0.3214944   3rd Qu.:-0.181331   3rd Qu.:-0.108057
##  Max.   : 3.1104345   Max.   : 5.514704   Max.   : 9.254284
##   state_Kansas       state_Kentucky     state_Louisiana      state_Maine
##  Min.   :-0.102072   Min.   :-0.06424   Min.   :-0.189456   Min.   :-0.069406
##  1st Qu.:-0.102072   1st Qu.:-0.06424   1st Qu.:-0.189456   1st Qu.:-0.069406
```

```
##  Median :-0.102072   Median :-0.06424   Median :-0.189456   Median :-0.069406
##  Mean   :-0.007157   Mean   : 0.01348   Mean   : 0.008921   Mean   : 0.000428
##  3rd Qu.:-0.102072   3rd Qu.:-0.06424   3rd Qu.:-0.189456   3rd Qu.:-0.069406
##  Max.   : 9.796892   Max.   :15.56746   Max.   : 5.278206   Max.   :14.407707
##  state_Maryland      state_Massachusetts state_Michigan     state_Minnesota
##  Min.   :-0.140626   Min.   :-0.087347   Min.   :-0.18341   Min.   :-0.005037
##  1st Qu.:-0.140626   1st Qu.:-0.087347   1st Qu.:-0.18341   1st Qu.:-0.005037
##  Median :-0.140626   Median :-0.087347   Median :-0.18341   Median :-0.005037
##  Mean   : 0.000581   Mean   :-0.001318   Mean   :-0.00946   Mean   :-0.005037
##  3rd Qu.:-0.140626   3rd Qu.:-0.087347   3rd Qu.:-0.18341   3rd Qu.:-0.005037
##  Max.   : 7.110959   Max.   :11.448450   Max.   : 5.45219   Max.   :-0.005037
##  state_Mississippi   state_Missouri      state_Montana      state_Nebraska
##  Min.   :-0.124107   Min.   :-0.184846   Min.   :-0.02666   Min.   :-0.003561
##  1st Qu.:-0.124107   1st Qu.:-0.184846   1st Qu.:-0.02666   1st Qu.:-0.003561
##  Median :-0.124107   Median :-0.184846   Median :-0.02666   Median :-0.003561
##  Mean   :-0.001108   Mean   :-0.007689   Mean   : 0.01000   Mean   :-0.003561
##  3rd Qu.:-0.124107   3rd Qu.:-0.184846   3rd Qu.:-0.02666   3rd Qu.:-0.003561
##  Max.   : 8.057487   Max.   : 5.409845   Max.   :37.50905   Max.   :-0.003561
##   state_Nevada      state_New_Hampshire state_New_Jersey   state_New_Mexico
##  Min.   :-0.09188   Min.   :-0.032266   Min.   :-0.15625   Min.   :-0.093823
##  1st Qu.:-0.09188   1st Qu.:-0.032266   1st Qu.:-0.15625   1st Qu.:-0.093823
##  Median :-0.09188   Median :-0.032266   Median :-0.15625   Median :-0.093823
##  Mean   :-0.01165   Mean   :-0.007476   Mean   : 0.00363   Mean   :-0.009819
##  3rd Qu.:-0.09188   3rd Qu.:-0.032266   3rd Qu.:-0.15625   3rd Qu.:-0.093823
##  Max.   :10.88396   Max.   :30.992131   Max.   : 6.40005   Max.   :10.658276
##  state_New_York      state_North_Carolina state_North_Dakota
##  Min.   :-0.219049   Min.   :-0.188123    Min.   :-0.055601
##  1st Qu.:-0.219049   1st Qu.:-0.188123    1st Qu.:-0.055601
##  Median :-0.219049   Median :-0.188123    Median :-0.055601
##  Mean   : 0.000544   Mean   : 0.004233    Mean   :-0.001145
```

```
##  3rd Qu.:-0.219049   3rd Qu.:-0.188123    3rd Qu.:-0.055601
##  Max.   : 4.565135   Max.   : 5.315606    Max.   :17.985019
##    state_Ohio       state_Oklahoma       state_Oregon       state_Pennsylvania
##  Min.   :-0.24185   Min.   :-0.128766   Min.   :-0.112186   Min.   :-0.099504
##  1st Qu.:-0.24185   1st Qu.:-0.128766   1st Qu.:-0.112186   1st Qu.:-0.099504
##  Median :-0.24185   Median :-0.128766   Median :-0.112186   Median :-0.099504
##  Mean   : 0.00022   Mean   :-0.001904   Mean   : 0.003739   Mean   :-0.003691
##  3rd Qu.:-0.24185   3rd Qu.:-0.128766   3rd Qu.:-0.112186   3rd Qu.:-0.099504
##  Max.   : 4.13473   Max.   : 7.765919   Max.   : 8.913645   Max.   :10.049748
##  state_Rhode_Island  state_South_Carolina state_South_Dakota
##  Min.   :-0.065224   Min.   :-0.175234    Min.   :-0.055601
##  1st Qu.:-0.065224   1st Qu.:-0.175234    1st Qu.:-0.055601
##  Median :-0.065224   Median :-0.175234    Median :-0.055601
##  Mean   : 0.002668   Mean   : 0.003007    Mean   : 0.007398
##  3rd Qu.:-0.065224   3rd Qu.:-0.175234    3rd Qu.:-0.055601
##  Max.   :15.331663   Max.   : 5.706569    Max.   :17.985019
##  state_Tennessee      state_Texas          state_Utah        state_Virginia
##  Min.   :-0.186090   Min.   :-0.209432   Min.   :-0.077852   Min.   :-0.15514
##  1st Qu.:-0.186090   1st Qu.:-0.209432   1st Qu.:-0.077852   1st Qu.:-0.15514
##  Median :-0.186090   Median :-0.209432   Median :-0.077852   Median :-0.15514
##  Mean   : 0.005264   Mean   :-0.008683   Mean   : 0.001693   Mean   : 0.01169
##  3rd Qu.:-0.186090   3rd Qu.:-0.209432   3rd Qu.:-0.077852   3rd Qu.:-0.15514
##  Max.   : 5.373684   Max.   : 4.774765   Max.   :12.844741   Max.   : 6.44588
##  state_Washington    state_West_Virginia state_Wisconsin     state_Wyoming
##  Min.   :-0.138358   Min.   :-0.087199   Min.   :-0.151584   Min.   :-0.04322
##  1st Qu.:-0.138358   1st Qu.:-0.087199   1st Qu.:-0.151584   1st Qu.:-0.04322
##  Median :-0.138358   Median :-0.087199   Median :-0.151584   Median :-0.04322
##  Mean   :-0.004299   Mean   : 0.002735   Mean   :-0.002997   Mean   : 0.00686
##  3rd Qu.:-0.138358   3rd Qu.:-0.087199   3rd Qu.:-0.151584   3rd Qu.:-0.04322
##  Max.   : 7.227518   Max.   :11.467822   Max.   : 6.596931   Max.   :23.13771
```

```
##    month_Jan           month_Feb         month_Mar            month_Apr
##  Min.   :-0.342776   Min.   :-0.3023   Min.   :-0.3282463   Min.   :-0.273329
##  1st Qu.:-0.342776   1st Qu.:-0.3023   1st Qu.:-0.3282463   1st Qu.:-0.273329
##  Median :-0.342776   Median :-0.3023   Median :-0.3282463   Median :-0.273329
##  Mean   : 0.005703   Mean   :-0.0159   Mean   : 0.0002255   Mean   : 0.009306
##  3rd Qu.:-0.342776   3rd Qu.:-0.3023   3rd Qu.:-0.3282463   3rd Qu.:-0.273329
##  Max.   : 2.917319   Max.   : 3.3077   Max.   : 3.0464543   Max.   : 3.658549
##    month_May          month_Jun           month_Jul           month_Aug
##  Min.   :-0.27968   Min.   :-0.268935   Min.   :-0.280471   Min.   :-0.310718
##  1st Qu.:-0.27968   1st Qu.:-0.268935   1st Qu.:-0.280471   1st Qu.:-0.310718
##  Median :-0.27968   Median :-0.268935   Median :-0.280471   Median :-0.310718
##  Mean   :-0.00393   Mean   :-0.002025   Mean   : 0.007022   Mean   :-0.003568
##  3rd Qu.:-0.27968   3rd Qu.:-0.268935   3rd Qu.:-0.280471   3rd Qu.:-0.310718
##  Max.   : 3.57546   Max.   : 3.718318   Max.   : 3.565391   Max.   : 3.218314
##    month_Sep           month_Oct           month_Nov
##  Min.   :-0.293926   Min.   :-0.307393   Min.   :-0.300145
##  1st Qu.:-0.293926   1st Qu.:-0.307393   1st Qu.:-0.300145
##  Median :-0.293926   Median :-0.307393   Median :-0.300145
##  Mean   : 0.006107   Mean   : 0.003866   Mean   :-0.009092
##  3rd Qu.:-0.293926   3rd Qu.:-0.307393   3rd Qu.:-0.300145
##  Max.   : 3.402178   Max.   : 3.253121   Max.   : 3.331677
##    month_Dec          weekday_Sun          weekday_Mon
##  Min.   :-0.323723   Min.   :-0.4218611   Min.   :-0.405841
##  1st Qu.:-0.323723   1st Qu.:-0.4218611   1st Qu.:-0.405841
##  Median :-0.323723   Median :-0.4218611   Median :-0.405841
##  Mean   : 0.002392   Mean   :-0.0001806   Mean   : 0.003175
##  3rd Qu.:-0.323723   3rd Qu.:-0.4218611   3rd Qu.:-0.405841
##  Max.   : 3.089023   Max.   : 2.3704186   Max.   : 2.463985
##   weekday_Tue         weekday_Wed         weekday_Thu          weekday_Fri
##  Min.   :-0.406710   Min.   :-0.41136   Min.   :-0.400868   Min.   :-0.398264
```

```
##  1st Qu.:-0.406710   1st Qu.:-0.41136   1st Qu.:-0.400868   1st Qu.:-0.398264
##  Median :-0.406710   Median :-0.41136   Median :-0.400868   Median :-0.398264
##  Mean   :-0.006036   Mean   :-0.00761   Mean   : 0.001086   Mean   :-0.001811
##  3rd Qu.:-0.406710   3rd Qu.:-0.41136   3rd Qu.:-0.400868   3rd Qu.:-0.398264
##  Max.   : 2.458723   Max.   : 2.43094   Max.   : 2.494555   Max.   : 2.510865
##   weekday_Sat       total_casualties  high_casualty
##  Min.   :-0.41262   Min.   : 0.0000   0:32906
##  1st Qu.:-0.41262   1st Qu.: 0.0000   1:  885
##  Median :-0.41262   Median : 1.0000
##  Mean   : 0.01132   Mean   : 0.7134
##  3rd Qu.:-0.41262   3rd Qu.: 1.0000
##  Max.   : 2.42350   Max.   :25.0000
```

```r
# Fitting a random forest model to predict high casualty incidents
# Exclude 'total_casualties' from predictors to prevent data leakage
rf_model <- randomForest(
  high_casualty ~ . - total_casualties,
  data = train_data_scaled,
  ntree = 100
)

# Predicting on the test set
# Remove 'high_casualty' from test data predictors
test_predictors <- test_data_scaled[, !(names(test_data_scaled) %in% c("high_casualty"))
predictions <- predict(rf_model, newdata = test_predictors)

# Evaluating the model with a confusion matrix
confusion_matrix <- table(test_data_scaled$high_casualty, predictions)
print(confusion_matrix)
```

```
##    predictions
```

```
##          0     1
##    0 32902     4
##    1   859    26
```

```r
# Calculating accuracy
accuracy <- sum(diag(confusion_matrix)) / sum(confusion_matrix)
print(paste("Accuracy:", round(accuracy, 4)))
```

```
## [1] "Accuracy: 0.9745"
```

```r
library(PRROC)
library(pROC)
library(ModelMetrics)

# Calculate Precision, Recall, and F1 Score
true_negatives <- confusion_matrix[1, 1]
false_positives <- confusion_matrix[1, 2]
false_negatives <- confusion_matrix[2, 1]
true_positives <- confusion_matrix[2, 2]

# Precision
precision <- true_positives / (true_positives + false_positives)

# Recall
recall <- true_positives / (true_positives + false_negatives)

# F1 Score
f1_score <- 2 * (precision * recall) / (precision + recall)

# Print the results
cat("Precision:", round(precision, 4), "\n")
```

```
## Precision: 0.8667
```

```r
cat("Recall:", round(recall, 4), "\n")
```

```
## Recall: 0.0294
```

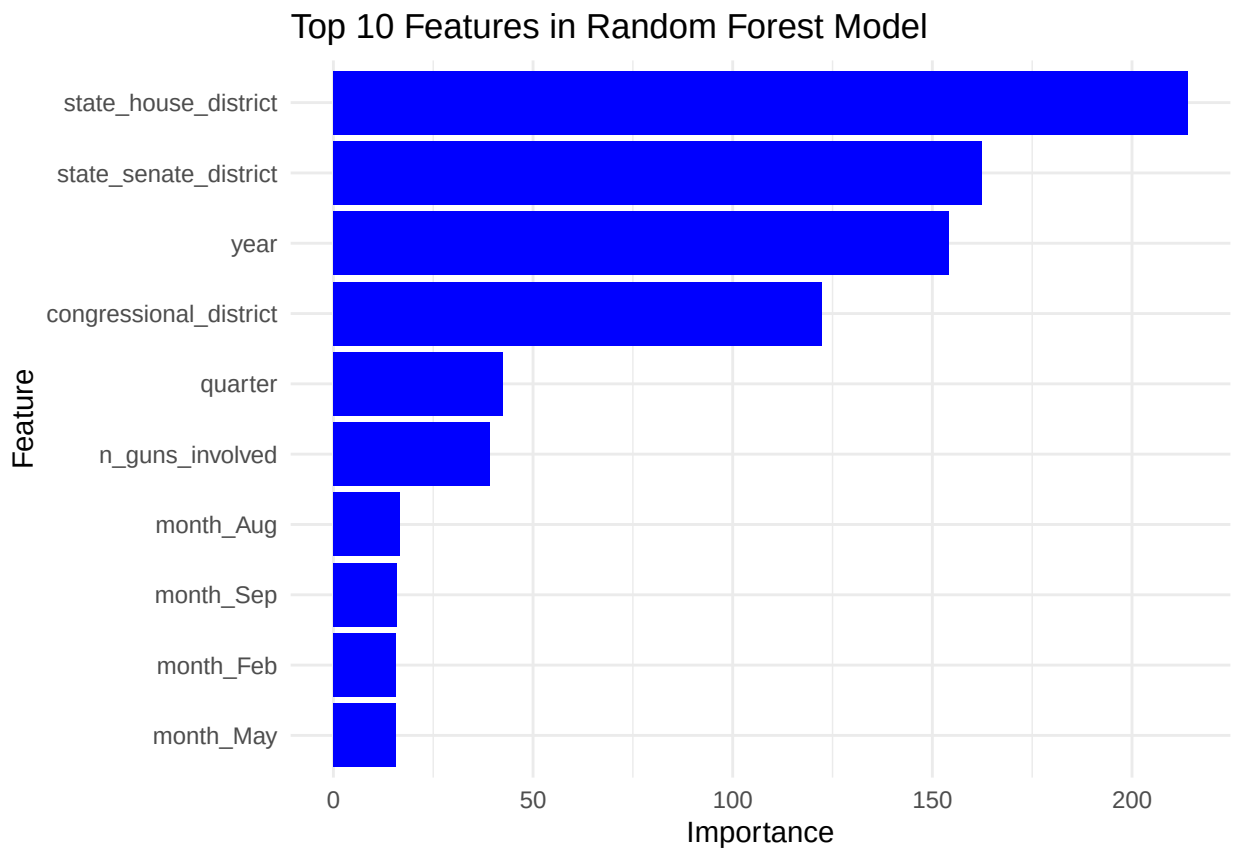
```r
cat("F1 Score:", round(f1_score, 4), "\n")
```

```
## F1 Score: 0.0568
```

```r
# Permutation Feature Importance - Displaying Top 10 Features
importance_values <- importance(rf_model)
importance_df <- data.frame(Feature = rownames(importance_values),
                            Importance = importance_values[, "MeanDecreaseGini"])

# Order the features by importance and select the top 10
top10_features <- importance_df[order(importance_df$Importance, decreasing = TRUE), ][1:
top10_features
```

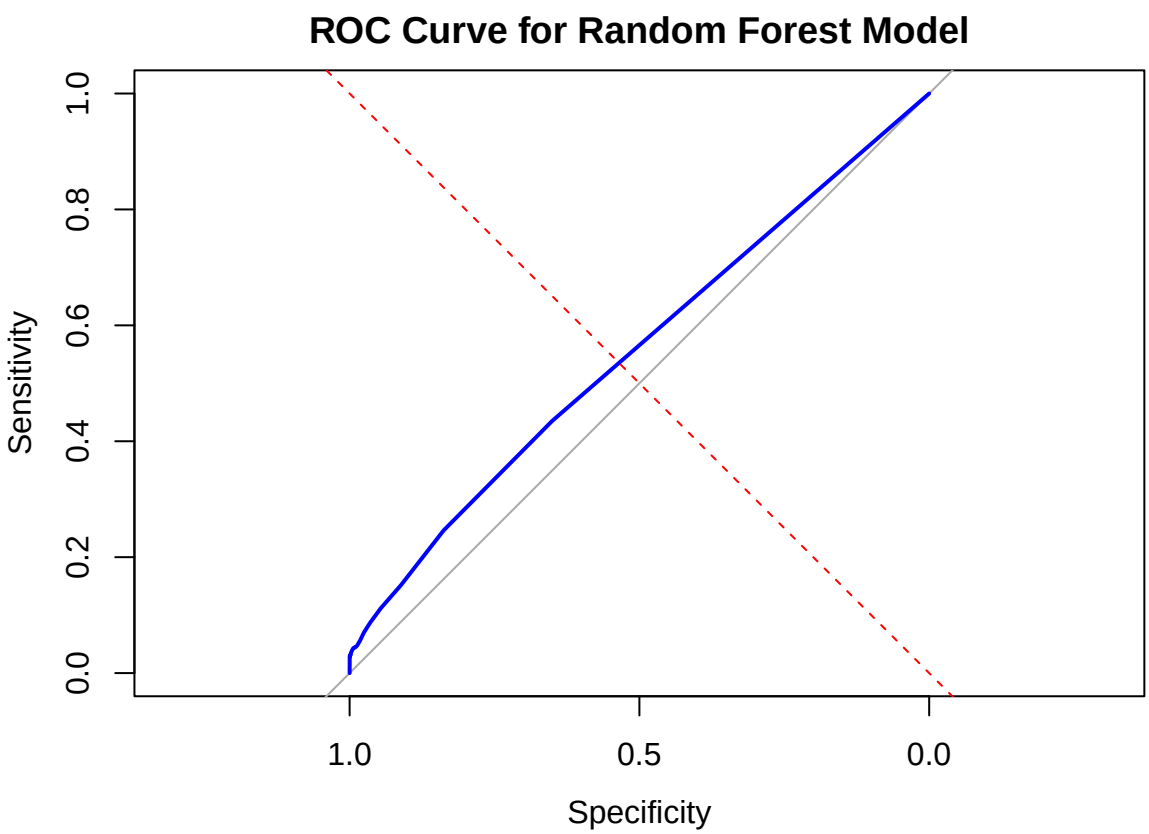
```
##                                      Feature Importance
## state_house_district     state_house_district  213.93786
## state_senate_district   state_senate_district  162.40548
## year                                     year  154.25073
## congressional_district congressional_district  122.33956
## quarter                               quarter   42.50228
## n_guns_involved               n_guns_involved   39.20299
## month_Aug                           month_Aug   16.61692
## month_Sep                           month_Sep   15.96157
## month_Feb                           month_Feb   15.69531
## month_May                           month_May   15.61685
```

```r
# Plot the top 10 variable importance
ggplot(top10_features, aes(x = reorder(Feature, Importance), y = Importance)) +
  geom_col(fill = "blue") +
  coord_flip() +
  labs(title = "Top 10 Features in Random Forest Model",
       x = "Feature",
       y = "Importance") +
  theme_minimal()
```




```r
# Predicting probabilities on the test set
test_probabilities <- predict(rf_model, newdata = test_data_scaled, type = "prob")

# Extracting the probabilities for the positive class (high casualty incidents)
positive_probabilities <- test_probabilities[, "1"]
```

```r
# Generating the ROC curve
roc_curve <- roc(test_data_scaled$high_casualty, positive_probabilities)

# Plotting the ROC curve
plot(roc_curve, col = "blue", main = "ROC Curve for Random Forest Model", lwd = 2)
abline(a = 0, b = 1, lty = 2, col = "red")
```




```r
# Set up cross-validation parameters
train_control <- trainControl(method = "cv", number = 10)
```

3.3.1 Insights:

1. **Top Features**: The random forest model identifies `state_house_district`, `state_senate_district`, and `year` as the most influential features, suggesting

political geography and time trends heavily impact incident severity.

2. **ROC Curve**: The ROC curve indicates limited discrimination capability, with sensitivity and specificity close to the random chance.

3. **Model Performance**: The model achieved high accuracy (~97%), but F1 score (0.057) is very low due to poor recall, indicating it struggles with correctly identifying high-casualty incidents.

4. **Cross-Validation**: Optimal model (mtry = 2) suggests low interpretability and variability across folds, with RMSE values indicating moderate error.

5. **Summary:** The model's predictive power is limited, likely due to class imbalance and overemphasis on geographic variables.

4 Results

4.1 Model Performance:

Accuracy: The model had an accuracy of `97.45%`.

Precision: `86.67%`

Recall: `2.94%`

F1 Score: `5.68%`

AUC-ROC: Indicates the model struggles to correctly identify high-casualty incidents due to limited sensitivity.

4.2 Feature Importance:

Key factors identified were `state_house_district`, `state_senate_district`, `year`, and `congressional_district`. This suggests that geographic and time factors significantly influence the severity of incidents.

4.3 ROC Curve and Calibration Curve:

These indicated the model's performance in distinguishing high-casualty incidents, though they also showed room for improvement.

5 New Insights

1. Incidents are more frequent among young adults and males.

2. Certain states and months have higher rates of severe incidents, confirming patterns seen in the EDA.

3. Locations like apartments and parks are common hotspots for incidents.

6 Connection to EDA and Initial Questions

1. **What factors predict severe incidents?** Factors like location type, time of year, and geographic district influence the severity of incidents.
2. **Are there times of the year when incidents are more likely?** Yes, certain months had higher incident rates like `Jan`, `July` and `Aug`.
3. **Which incidents are most likely to have multiple casualties?** Incidents in specific locations like `apartments`, `parks`, `Walmart` and `7-11` showed a higher likelihood of multiple casualties.
4. **How do local factors relate to high incident rates in certain states?** States with large cities like `Illinois`, `California` show higher incident rates, suggesting that urban areas may need more focused interventions.

7 Interpretations

7.1 Implications of Results

For Policymakers: Focused efforts should be directed toward states and months with higher incident rates. Increasing safety measures in specific locations like `apartments` and `parks` could also reduce gun violence.

For Law Enforcement: Monitoring patterns over time and geographic areas can help them prepare better and allocate resources effectively.

8 Model Comparison

1. Logistic Regression: Failed to make meaningful predictions due to severe class imbalance.

2. LightGBM: Achieved perfect accuracy (AUC = 1) on both training and testing data. Key Issue: Likely overfitted, as this performance seems unrealistic and may not generalize well.

3 K-Nearest Neighbors (KNN): Same results with LightGbm, and also struggled with class imbalance and high-dimensional data.

9 Recommendations & Conclusions

9.1 Recommendations

1. Policy Actions: Increase security in high-risk locations like `apartments`, `parks` and during high-incident months.

9.2 Next Steps

1. Data Collection: To improve model accuracy, I can incorporate additional data, such as socioeconomic factors (e.g., poverty rates, education levels) and gun ownership statistics.

These variables could offer deeper insights into factors influencing gun violence, allowing for more accurate predictions and targeted recommendations.

2. Model Improvement: To improve model performance, I can address class imbalance using techniques like SMOTE, test robust models (e.g., XGBoost) with class weighting, and focus on recall-oriented metrics like Precision and F1 Score. Additionally, parameter tuning through cross-validation will help achieve a balanced and generalized model.

9.3 Final Thoughts

The findings connect well with the initial questions, showing clear patterns in gun violence that vary by time, location, and participant demographics. By focusing on these high-risk areas and times, authorities can develop better strategies to reduce severe incidents.

10 Summary

In this analysis of gun violence incidents, I aimed to find the factors and patterns that predict severe cases. I tested different machine learning models, like Logistic Regression, K-Nearest Neighbors (KNN), LightGBM, and Random Forest, to identify high-casualty incidents. Each model gave unique insights but also had challenges. For example, Logistic Regression and KNN didn't perform well due to the imbalanced data, leading to poor classification. LightGBM showed high precision but overfitted the data, which made it less reliable for new data.

The Random Forest model worked best, balancing accuracy and feature importance. It highlighted key factors, such as specific months and states with higher risk. However, the Random Forest model could still be better. Adjusting parameters like the number of trees and using techniques to balance data, such as SMOTE (Synthetic Minority Over-sampling Technique), might improve its recall and overall strength.

These insights are valuable for policymakers and law enforcement, helping them focus resources during high-risk times and in critical places. Future studies could make the model

even more accurate by adding other variables, like socioeconomic factors, to better understand what causes these incidents.

11 References

1. [Gun Violence Dataset] (https://www.kaggle.com/datasets/jameslko/gun-violence-data)

2. [Gun Violence Data - Discussions from Kaggle] (https://www.kaggle.com/datasets/jameslko/gun-violence-data/discussion)

3. [Gun Violence Data - Dataset Notebooks from Kaggle] (https://www.kaggle.com/datasets/jameslko/gun-violence-data/code)

4. [Data Cleaning and Manipulation] (https://nceas.github.io/sasap-training/materials/reproducible_research_in_r_fairbanks/data-cleaning-and-manipulation.html)